University of Cape Town

Honours Project

A simulation study exploring the design of camera trap surveys for Spatial capture-recapture models

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Abstract

Motion-triggered camera traps have become essential tools because they can "capture" certain species that are difficult to observe otherwise. When these species can be uniquely identified, these data are ideal for Capture-recapture (CR) models. Spatial capture-recapture (SCR) models further use information from spatial locations of repeat encounters of individuals to estimate population density. By incorporating spatial data, SCR models enhance traditional non-spatial CR methods and have become the standard approach for modelling animal populations. Despite their potential, existing guidelines for trap placement remain largely heuristic and untested across diverse ecological scenarios. This paper investigates the optimisation of survey designs for SCR models, concentrating on three reference case scenarios: leopards, roe deer, and Sumatran striped rabbits. Each species exhibits distinct ecological characteristics, which are critical for assessing the effectiveness of different SCR design methodologies. The performance of various approaches in generating SCR designs across these case studies is evaluated through simulation studies, focusing on their impact on key parameters such as capture rates and the precision of density estimates. Findings indicate that the case study characteristics affect the precision and accuracy of density estimates, particularly when density varies spatially. This research contributes to developing more effective wildlife monitoring strategies, enabling conservation practitioners to make informed decisions regarding trap design and placement for enhanced data collection and population assessment.

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Contents

1 Introduction **1**
1.1 Background of Spatial Capture-Recapture 1
1.2 Problem Statement 2
1.3 Research Objectives 2

2 Methodology **4**
2.1 SCR Model Formulation and Implementation 4
2.2 Design Optimisation 6
2.3 Design Constraints 9
2.4 Case Studies 12
2.5 Generation of Designs 13
2.6 Simulation-Based Evaluation 14

3 Results **16**
3.1 Designs 16
3.2 Simulated Data Quality 18
3.3 Evaluation of Density Estimates 24
3.3.1 Homogeneous Density 24
3.3.2 Inhomogeneous Density 27

4 Discussion **31**

5 Conclusion **35**

A Simulated Data Quality **36**

B Evaluation of Density Estimates **40**

C R-Code **44**

List of Figures

2.1 Half-Normal detection function 6
2.2 Spatial variation of density 11
2.3 All possible trap locations 13
2.4 Examples of realisations of the regular grid 14
3.1 Proposed optimal designs for the leopard 16
3.2 Proposed optimal designs for the deer 17
3.3 Proposed optimal designs for the rabbit 17
3.4 Expected and observed detection data for the leopard 20
3.5 Expected and observed detection data for the deer 22
3.6 Expected and observed detection data for the rabbit 23
3.7 Average RB(%) and RSE(%) for homogeneous density 26
3.8 Average RB(%) and RSE(%) for the gradient in density 29
3.9 Average RB(%) and RSE(%) for conical density 30
A.1 Expected and observed detection data over the conical density surface 39
B.1 Boxplots of select homogeneous density estimates for the rabbit . . . 40
B.2 Boxplots of select conical density estimates for a leopard 40

List of Tables

2.1 Case study parameters 12
A.1 Average expected and observed detection data for the leopard 36
A.2 Average expected and observed detection data for the deer 37
A.3 Average expected and observed detection data for the rabbit 38
B.1 Performance metrics for the leopard 41
B.2 Performance metrics for the deer 42
B.3 Performance metrics for the rabbit 43

1 Introduction

1.1 Background of Spatial Capture-Recapture

Understanding how many individuals of a species live in a given area is vital for wildlife conservation and management efforts. Capture-recapture (CR) models are commonly used to estimate abundance, the size of the population, by repeatedly sampling individuals in a population with replacement (Sun et al., 2014). Spatial capture-recapture (SCR) models take this approach a step further by incorporating the locations where individuals are detected. This added spatial data helps estimate both population size and density, accounting for variability in detection probabilities among individuals based on their proximity to sampling devices (Sun et al., 2014). Population density, which measures the number of individuals per unit area, is often more useful than abundance because it allows for better comparisons across different areas and time periods. SCR models are now widely regarded as the standard for estimating population density, enhancing traditional CR methods (Borchers & Efford, 2008).

The development of detectors, such as motion-triggered camera traps and hair snares, allows species that were previously difficult to observe without physical capture to be "captured" using unique identifiers. Individuals can be identified through genetic analysis from samples from hair snare traps. However, camera traps are especially useful for more elusive species as they simplify individual identification by matching unique characteristics to specific individuals. For example, each leopard has its own unique rosette patterns, which enable modelling using SCR techniques.

All spatial capture-recapture surveys must determine the optimal placement of detectors to effectively obtain population parameter estimates (Durbach et al., 2021). Camera trap surveys are costly and time-intensive, particularly in remote and rugged terrain. Despite this, little research has focused on how best to use limited resources when deciding where to place the detectors, also known as designing the trap array (Dupont et al., 2021).

It is the industry norm to employ a regular grid design, typically a 2σ or $2km$ grid of detectors, where traps are spaced at regular intervals of 2σ metres or $2km$ (Durbach et al., 2021). When basic guidelines are adhered to, these designs work sufficiently well and are expected to provide unbiased density estimates with relatively good precision under various conditions (Dupont et al., 2021). However, in some survey areas, regular spacing may not be practical. Therefore, even if regular spacing is feasible, alternative designs could lead to better estimates (Dupont et al., 2021).

1.2 Problem Statement

Camera trap surveys are essential for providing data to estimate parameters such as density using SCR models. Despite their widespread application, optimal trap placement guidelines for reliable estimates remain largely heuristic and untested, particularly across species with differing movement patterns and habitat use. An inefficient trap layout can result in poor data leading to biased and imprecise estimates, which hinders effective wildlife monitoring and conservation efforts, especially for species that are rare or elusive. This study focuses on using simulation to evaluate the performance of various trap design optimisation methods across three different species, each exhibiting distinct ecological characteristics. By comparing different design optimisation approaches, this paper seeks to develop near-optimal survey designs for species with varying characteristics, thus providing a framework from which ecologists can make design choices.

1.3 Research Objectives

Currently, there are different optimisation approaches which yield proposed optimal trap designs. However, these approaches have not been tested for various case studies which would reveal if the methods perform equally well across different combinations of parameters. This paper evaluates how the methods perform for a spectrum of species with varying characteristics, assuming a closed population. The focus is on three case studies, namely the leopard, the Sumatran striped rabbit, and the roe deer.

This paper uses a genetic algorithm (GA) with different design criteria outlined by Dupont et al. (2021) and Durbach et al. (2021) to develop designs for the various case studies and evaluate which approaches yield the best design specific to each biological setup.

The objective is to find near-optimal designs for each different set of characteristics that ecologists can then use as a framework when conducting camera trap surveys. One approach may work better than another for a wide-ranging, low-density animal, and this information is more helpful than the broad, heuristic guidelines that currently exist for designs (Sun et al., 2014). This will enable conservation researchers and scientists to create effective sampling surveys for monitoring wildlife. Consequently, these designs will yield a higher level of data, resulting in more precise estimates of key population parameters.

At first, very large study areas will be used to ensure the optimisation algorithms are not constrained when choosing detector placements. Thereafter, smaller areas more representative of real-world study areas will be explored. Additionally, constant and non-constant densities are explored to ensure the robustness of the research. Designs are first evaluated using homogeneous density, with individuals occurring equally across the study area. In reality, animals are unevenly distributed, and accounting for this may show that certain design approaches perform better when density varies in space, whilst others don't. Therefore, inhomogeneous density is included as a feature of the simulation study.

2 Methodology

2.1 SCR Model Formulation and Implementation

Initially, density was estimated using an inverse prediction method, which, although there has been some success using this method, is limited in terms of model selection and the inclusion of covariates (Efford et al., 2004). Borchers and Efford (2008) later developed the maximum likelihood SCR framework, integrating individual encounter histories with the spatial distribution of detectors and unobserved animal activity centres $\boldsymbol{s_i}$, defined by a pair of coordinates (Sollmann et al., 2012). The activity centre, sometimes interpreted as the home range centre, represents the point where an individual spends the most time or is most active. All possible activity centres, $\boldsymbol{s}$, are defined within a state-space $\boldsymbol{S}$, which encompasses the trapping grid and is large enough to include all individuals that could interact with it (Sollmann et al., 2012).

The SCR models introduced a spatial component previously absent in likelihood-based capture-recapture models (Borchers & Efford, 2008). Without this spatial component, robust density estimates are unattainable; furthermore, point estimators using the spatial models are nearly unbiased (Borchers & Efford, 2008). SCR models rely on two primary processes: the state and observation models. The state model describes the likely distribution of animal locations by assigning probabilities to where each animal is most likely to be found within a defined area (Borchers, 2010). The observation model, in turn, describes how detections are made, given the animals' locations (Borchers, 2010). The observation model determines the probability of capturing an animal based on the distance between an individual's activity centre and a particular detector.

The likelihood is the joint distribution of the number of captured animals n, and their capture histories $\boldsymbol{\omega}_1, \ldots, \boldsymbol{\omega}_n$. According to the conditional probability theorem, this can be expressed in terms of the marginal distribution of n and the conditional distribution of $\boldsymbol{\omega}_1, \ldots, \boldsymbol{\omega}_n$ given n as

$$L(\boldsymbol{\phi}, \boldsymbol{\theta} | n, \boldsymbol{\omega}_1, \ldots, \boldsymbol{\omega}_n) = Pr(n|\boldsymbol{\phi}, \boldsymbol{\theta}) Pr(\boldsymbol{\omega}_1, \ldots, \boldsymbol{\omega}_n | n, \boldsymbol{\theta}, \boldsymbol{\phi}) \tag{2.1}$$

Where $\boldsymbol{\theta}$ is the vector of observation model parameters and $\boldsymbol{\phi}$ is a vector of parameters of the state model governing animal density and distribution (Borchers & Efford, 2008). However, the exact location of the activity centres is usually unknown. To account for this uncertainty, all possible locations of the activity centres are

integrated when estimating the overall likelihood of the observed capture data (Borchers & Efford, 2008). Equation 2.2b shows the probability of observing capture history $\boldsymbol{\omega}_i$ for individual i, given that it was captured.

$$Pr(\boldsymbol{\omega}_1, \ldots, \boldsymbol{\omega}_n | n, \boldsymbol{\theta}, \boldsymbol{\phi}) = Pr(\boldsymbol{\omega}_1, \ldots, \boldsymbol{\omega}_n | \boldsymbol{\omega}_1 > 0, \ldots, \boldsymbol{\omega}_n > 0, \boldsymbol{\theta}, \boldsymbol{\phi}) \tag{2.2a}$$

$$Pr(\boldsymbol{\omega}_i | \boldsymbol{\omega}_{i.} > 0, \boldsymbol{\theta}, \boldsymbol{\phi}) = \int Pr(\boldsymbol{\omega}_i | \boldsymbol{\omega}_{i.} > 0, \boldsymbol{\theta}, \boldsymbol{S}) f(\boldsymbol{S} | \boldsymbol{\omega}_{i.} > 0, \boldsymbol{\theta}, \boldsymbol{\phi}) d\boldsymbol{S} \tag{2.2b}$$

Equation 2.1 demonstrates that the likelihood is the combination of the state process, the probability of observing the n captured animals, and the conditional probability of the capture histories on that n (Equation 2.2b). The method used to calculate the probability of capture histories depends on the type of detector employed; if one were observing multiple sampling periods, one could use any detector type. For example, a binary detector with a binomial probability function. This study focused on a single observation period, utilising count proximity detectors, common with camera traps, which record detection frequencies. The parameter λ, from a Poisson probability function, determines the intensity of the spatial point process, which distributes the activity centres.

The encounter rate for an individual, i, in each of $j = 1, \ldots, \mathcal{J}$ traps, y_{ij}, is a Poisson random variable with rate parameter λ_{ij} (Sun et al., 2014). The parameter λ_{ij} represents the capture rate in the detection function, indicating the average expected number of detections for an individual based on the distance from its activity centre to a particular trap j, obtained from the state process. It reflects the expected number of captures at a given distance from the activity centre. The detection function has various forms but is typically modelled as a half-normal distribution, seen in Figure 2.1 below, where σ is the spatial scale parameter representing the decay rate of the detection function. It corresponds to an individual's range of movement from its activity centre (Sun et al., 2014).

The encounter rate at a distance of zero from trap j, λ_{0j}, represents the expected capture rate when a trap is optimally placed on an individual's activity centre. λ_{0j}, therefore, corresponds to the y-intercept of the half-normal detection function in Figure 2.1, signifying the highest encounter rate, as expected.

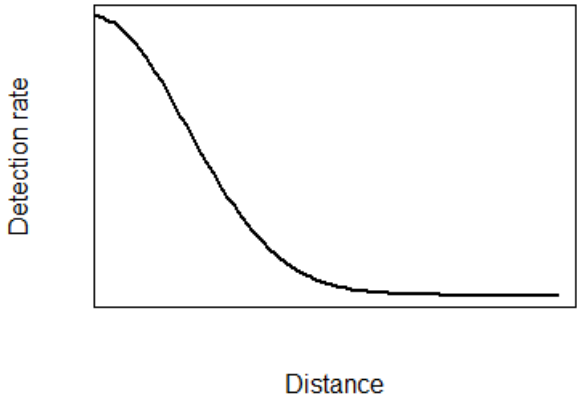

Figure 2.1: Visualisation of a Half-Normal detection function (λ_{0j}, σ).

2.2 Design Optimisation

When devising camera trap designs, it is crucial to balance the trade-off between maximising the number of animals captured (denoted as n) and ensuring a sufficient number of recaptures (denoted as r) at a useful spatial resolution. If the traps are spread far apart, they may capture many individuals initially, but the chances of recapturing those individuals at different traps are diminished. Conversely, placing traps close together may yield high recapture rates but limit the capture of new individuals or result in redundant recaptures of the same individuals at neighbouring traps. Detection of the same animal at different detectors placed very close together provides little information about the shape of the detection function. Detectors being close together reduces spatial coverage and leads to many recaptures at relatively uninformative fractions of σ (Durbach et al., 2021). This is not ideal for SCR as it provides little to no information on an individual's movement range.

With this in mind, Efford and Boulanger (2019) argue that because SCR estimators are generally unbiased, the focus of study design should shift from minimising bias to maximising precision, which is essential for improving accuracy. Traditionally, assessing variance and bias required computationally intensive simulations, where each potential design was evaluated using numerous simulated datasets, making the process slow and subjective. Thus, it is crucial to employ more efficient design strategies that balance computational feasibility with the need to generate high-quality, reliable data (Durbach et al., 2021).

To address this, Efford and Boulanger (2019) introduced a new approach to assess the optimality of designs, aiming to improve precision beyond the commonly referenced target of a relative standard error (RSE) below 20%. While an RSE $< 20\%$ provides

sufficient capacity to detect large population changes, their approach encourages more ambitious designs that not only improve the precision of estimate but also enable the detection of smaller yet meaningful changes (Efford & Boulanger. 2019). Since wildlife surveys are often interested in density, it is logical for candidate designs to consider minimising the mean square error of density estimates as an evaluation criterion. This corresponds to the square of the bias plus the variance, but SCR estimators are known to be unbiased (Durbach et al., 2021). Thus, optimal designs can be assessed by maximising the precision of density estimators, which is the same as minimising the coefficient of variation in the density estimator, $\mathrm{CV}(\widehat{D})$ (Durbach et al., 2021). Using this, Efford and Boulanger (2019) applied the following simple approximation to optimise detector locations efficiently:

$$\mathrm{CV}(\widehat{D}) \approx \frac{1}{\sqrt{\min\{E(n), E(r)\}}} \tag{2.3}$$

This equation uses the expected number of captures, $E(n)$, and the expected number of recaptures, $E(r)$, to approximate the precision of density estimates. This approximation was pivotal as it allowed for a quick comparison of candidate designs because both $E(n)$ and $E(r)$ can be computed easily using numerical integration, eliminating the need for computationally expensive simulations. This enabled researchers to easily understand how varying the configurations of detectors could influence the precision of the density estimate.

Parallel to this, Dupont et al. (2021) introduced a genetic algorithm (GA) that produces near-optimal sampling designs by minimising a criteria-based objective function. The GA is used to create a design that minimises specific criteria, namely $Q_{\bar{p}} = -\bar{p}(X)$, $Q_{\bar{p}_m} = -\bar{p}_m(X)$ and $Q_{\bar{p}_b} = Q_{\bar{p}} + Q_{\bar{p}_m}$. $\bar{p}(X)$ represents the marginal probability of encounter, and $\bar{p}_m(X)$ is the probability that an individual is detected at two or more detectors (Dupont et al., 2021). Conceptually, optimising with respect to $Q_{\bar{p}}$, the space-filling objective function, tends to spread traps out across the study area, aiming to maximise the expected sample size n. In contrast, $Q_{\bar{p}_m}$ tends to place traps in clusters to increase the number of recaptures. The third criterion, $Q_{\bar{p}_b}$, attempts to strike a balance between spreading traps and clustering them (Dupont et al., 2021).

Initially, the genetic algorithm generates a population of several hundred random designs. These designs are then evaluated based on the chosen criterion, with favourable designs being preferentially selected with replacement. Subsequently, selected designs are randomly paired, and each pair undergoes a process where a

portion of their locations is exchanged to produce two new designs. Following this exchange, a final mutation step introduces a small probability of replacing each location with a randomly selected one. This process is repeated for a fixed number of iterations or until convergence is reached, outputting a proposed optimal design. (Dupont et al., 2021)

The results from this work show that the $Q_{\bar{p}}$ and $Q_{\bar{p}_b}$ criteria do not perform well when too few traps are used, as they cannot produce meaningful spatial recaptures. On the other hand, the $Q_{\bar{p}_m}$ criteria performed better in most cases, generating clustered designs due to the inherent trade-off between n and r (Dupont et al., 2021).

Durbach et al. (2021) then combined the approximation of the precision of density estimates, CV($\widehat{D}$), proposed by Efford and Boulanger (2019), with the optimisation techniques used by Dupont et al. (2021), using CV($\widehat{D}$) as the design criterion in place of the original criteria applied by Dupont et al. (2021). Their goal was to illustrate how optimisation techniques can be used to identify optimal detector placements which don't conform to a regular layout and that maximise the approximate precision of density estimates (Durbach et al., 2021). The detector layouts produced when using the approximation as the optimisation criterion are called min(n,r) designs. They demonstrate the balance between widely spaced traps, which yield numerous initial captures but diminish recapture rates, and clustered traps, which maintain consistent recapture rates across traps (Durbach et al., 2021). They also expanded on the work of Efford and Boulanger (2019) by developing designs which incorporate inhomogeneous density to show that the CV($\widehat{D}$) approximation remains accurate even when density varies spatially (Durbach et al., 2021).

In order to implement the genetic algorithm, also referred to as the optimisation approach, a specification of all potential trap locations and a mask must be provided. All potential trap locations are usually inputted in the form of a mesh of regularly spaced points over the study area. Based on the mesh, a mask of the study area is created to define the space where potential activity centres could exist, ensuring it includes all regions with a non-zero probability of being exposed to the traps. This restricts trap placement to suitable areas for the survey. The mask includes a buffer around the mesh to account for animal movement outside the immediate trapping area. The purpose of a buffer is to ensure that the detection process includes individuals whose activity centres might be outside the camera trap array but can still enter the detection area. It is important to determine an optimal resolution for the mask, as the process of numerically integrating out the activity

centres depends on the mask's resolution. A too coarse resolution of cells will result in biased estimates, but a too fine mask has a much higher computational cost as there are more cells.

Similarly, whilst smaller spacing between possible trap locations offers more flexibility for optimisation, capturing the same individual at very short distances adds little value, as discussed previously. Thus, the coarseness of the mesh of potential trap locations is vital for criteria such as min(n, r) and $Q_{\bar{p}_m}$, which treat all recaptures as equivalent. This is because placing detectors as close together as the spacing allows maximises the total number of recaptures (Durbach et al., 2021). However, placing detectors very close together also reduces the number of unique individuals caught and increases the number of uninformative recaptures, inflating CV($\widehat{D}$) (Durbach et al., 2021). Once an appropriate mesh spacing of $\frac{2}{3}\sigma$ was used, their approach produced designs which are competitive with regular grid designs and, in some cases, outperform them, especially when animals are unevenly distributed (Durbach et al., 2021). In comparison to the regular grid, min(n, r) designs can be used in study areas of any shape, offering an advantage of flexibility, and are likely easier and more cost-effective to implement due to the closer placement of detectors (Durbach et al., 2021).

2.3 Design Constraints

This paper's primary objective was to assess the effectiveness of SCR designs across various biologically realistic scenarios, gaining a broader understanding of how the following design constraints influence performance: the size of the study area - as performance of SCR models may be limited if small sampling arrays, relative to individual movement, are used (Sollmann et al., 2012), the number of traps available for the design, the density pattern - defined as the way in which density was varied spatially, and species with varying movement ranges, densities, and ecological traits. The parameters used in the figures of the next sections pertain to the scenario involving a large grid with 40 detectors and a homogeneous density distribution for leopards with a density (D) of 7 leopard per $100km^2$.

The size of the area:
Due to the space-filling nature of certain algorithms, it was prudent to assess the performance of the models over a large area, albeit larger than many feasible study areas, to ensure robust results. Therefore, the mask size, including the buffer, was initially set to be a large grid of 30σ in each direction.

A smaller mask was also evaluated based on more realistic study areas from which the case study parameters were derived. Care was taken to ensure the mask was large enough to cover the widely used regular grid - this paper used a regular grid of detectors with 2σ spacing. The smaller mask was therefore chosen to be 18σ by 24σ.

The number of traps:
This paper initially examined the effectiveness of using a grid with 40 detectors. Prior research indicates this approach should produce relatively unbiased results (Durbach et al., 2021). As a result, the impact of using fewer detectors (20) was evaluated, rather than exploring the use of additional detectors, to accommodate the limitations faced by resource-constrained researchers.

Density patterns:
This paper explored density patterns using artificial areas; therefore, it is plausible that different findings may have been obtained if more realistic surfaces had been used. This paper explored three variations of density by creating different density surfaces as covariates in the mask. These synthetic covariates could be substituted for appropriate covariates of interest for specific research, as current variations in density are not necessarily biologically plausible.

Three types of density surfaces, based on research by Durbach et al. (2021), were analysed;

i. The uniform distribution of activity centres. Density D, is a constant parameter for this surface informed by the respective case studies.

ii. Density increases with latitude and longitude, resulting in the highest densities at the northeast edge of the habitat mask, which is in the buffer region and inaccessible to detectors. Therefore, density is expected to be underestimated, and the estimate is biased (Durbach et al., 2021). This surface is observed on the left grid of Figure 2.2 below as the gradient in density and was calculated as follows:

$$D_1 \propto \exp\left(-\sqrt{(x-\bar{x})^2 + (y-\bar{y})^2}\right) \tag{2.4}$$

where x and y are default coordinates in the mask. Therefore it is possible to model spatial density without a covariate.

iii. Density is concentrated at the centre of the survey area, gradually decreasing as distance from the centre increases. This is observed on the right grid of Figure 2.2 below as the conical density and was calculated as follows:

$$D_2 \propto \exp\left(-\sqrt{xy}\right) \tag{2.5}$$

The interaction between x and y offers a flexible density surface.

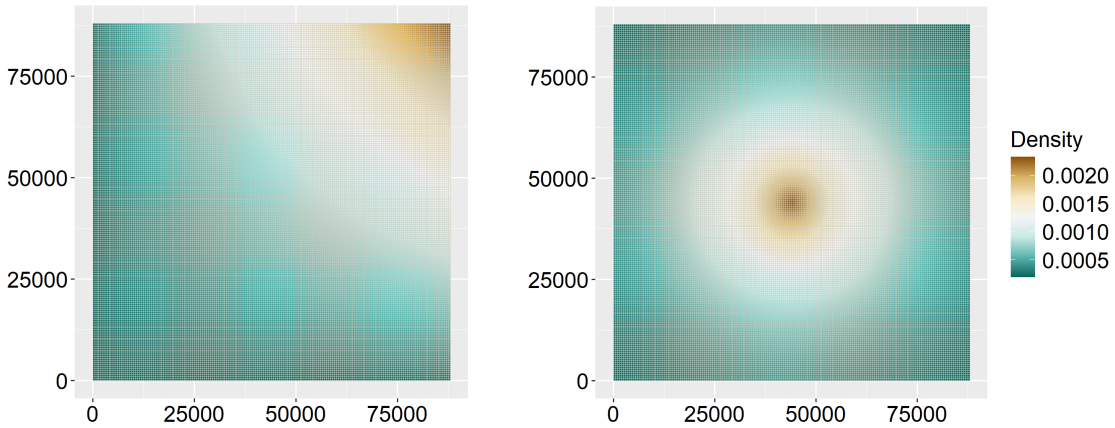

Figure 2.2: Spatial variation of density across the study area, displaying a gradient in density (left) and conical density (right). The horizontal and vertical coordinates are in meters. Density surface Equations (2.4) and (2.5) were standardised such that the mean of the densities equals the relative uniform density, $D = 0.0007$ leopards.

Species:

This paper analyses three case studies to evaluate the effectiveness of different design methodologies in estimating model and detection parameters, with a focus on the precision of density estimates for various species. The species studied — leopards (*Panthera pardus*), roe deer (*Capreolus capreolus*), and Sumatran striped rabbits (*Nesolagus netscheri*) — were selected because individuals can be uniquely identified from photographs. They differ significantly in key ecological traits such as movement range and population density, which are critical for assessing the design methods. Parameters for the simulation data were derived from species-specific studies to ensure that the estimates are relevant and realistic.

It is possible to model detection parameters across multiple occasions; however, using a single occasion is common with count models when there is no interest in modelling variation in detections over time. Therefore, in this study, the survey period was a 45-day period representing one occasion using count proximity detectors. The

encounter rates represent the expected frequency of captures over this period and baseline encounter rates were scaled to reflect the 45-day period.

2.4 Case Studies

Leopards typically occur at low densities, range over large areas, and are elusive, solitary, and nocturnal, making them challenging to detect and monitor reliably (Balme et al., 2019). They are identifiable by the unique rosette patterns on their coats. The leopard simulation parameters in this study were based on a camera-trapping survey conducted by Panthera in Pilanesberg, 2024 (Woodgate et al., 2024).

In contrast, roe deer are social animals that occur at high densities and have a moderate range of movement, making them more detectable than the elusive leopards (Hinojo et al., 2022). The simulation parameters for roe deer were drawn from Hinojo et al. (2022) which estimated male roe deer density in Versiox, Switzerland, using camera trapping. Male roe deer can be uniquely identified by the shape, length, and curvature of their antlers.

The Sumatran striped rabbit is an ideal subject for SCR studies due to its distinctive stripe fur pattern and the challenges in monitoring this species due to its rarity and the remote, densely forested habitat it occupies (McCarthy et al., 2019). Similarly to the leopard, it occurs at low densities; however, due to its small size, it has a small range of movement. Little research has been done on the Sumatran striped rabbit, and it is classified as data-deficient by the IUCN (McCarthy et al., 2019), therefore parameters for this species were based on SCR analyses of a similar lagomorph, the snowshoe hare (*Lepus americanus*; Kenney et al., 2023). The parameters used for all simulations are detailed in Table 2.1 below. For convenience, each case study will be referred to by the respective labels: leopard, deer and rabbit.

Table 2.1: Table of the population parameters for each case study. λ_0 is the encounter rate at a distance of zero from all traps. It is scaled according to the 45-day occasion. σ is the animal's range of movement (in meters). D is the density of the species per $10km^2$.

Species	λ_0	σ (m)	D per $10km^2$
Leopard	0.9	3000	0.7
Deer	12.6	336.4	39
Rabbit	6.75	80	180

2.5 Generation of Designs

Design generation was done using the SCR model parameters from the case studies, a mesh of all possible trap locations, a mask of the study area, and the number of traps available, which were all inputted into the GA. As previously explained, the coarseness of the mesh of potential trap locations is important to ensure accurate results; therefore, in keeping with the work done by Durbach et al. (2021), a conservative spacing of $\frac{2}{3}\sigma$ between possible trap locations was used. Similarly, for the mask, a resolution of $\frac{1}{6}\sigma$ was found to be optimal, along with a buffer of 3σ.

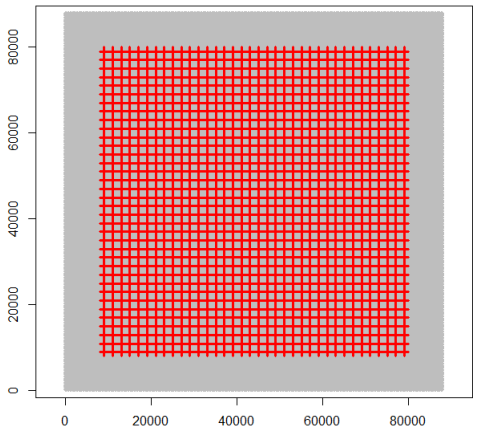

Figure 2.3: Mesh of all possible trap locations (indicated by the red crosses) superimposed on the mask with the surrounding buffer. The horizontal and vertical coordinates are in meters.

Once the mesh of all possible trap locations and corresponding mask had been produced, as seen in Figure 2.3, the optimisation algorithm was implemented. The balancing objective, min(n,r), which uses the approximation of CV($\widehat{D}$), and the clustering objective, $Q_{\bar{p}_m}$, were chosen to be the two optimisation criteria. The other two criteria suggested by Dupont et al. (2021) were not used as they are less effective. To ensure algorithm convergence, 50 generations of the GA were used (Durbach et al., 2021). The GA is also a stochastic process, and therefore, 100 designs were generated for each criterion, which were then evaluated through simulation, as described in the next section.

In order to assess the optimisation approaches, a base case was needed for comparison, so the industry norm of a regularly spaced 2σ grid was chosen. Only one realisation of the regular grid, seen on the left in Figure 2.4, was necessary when using homogeneous

density. This is because the density was constant over the entire study area, so it did not matter where the grid was placed. However, when using either of the inhomogeneous density surfaces, the placement of the regular grid needed to be randomised to account for the varying density pattern. In keeping with the approach used for the proposed optimal designs, 100 random regular grids were generated for each scenario; one such example can be seen in Figure 2.4 on the right. The k-nearest neighbours (KNN) algorithm generated these designs, meaning the resulting grids may not necessarily be rectangular. Nevertheless, they conform to the standard of regular spacing.

The min(n,r) designs, clustering $Q_{\bar{p}_m}$ designs, and regular grids were generated with the different combinations of design constraints using homogeneous density, then a gradient in density and finally, a conical density.

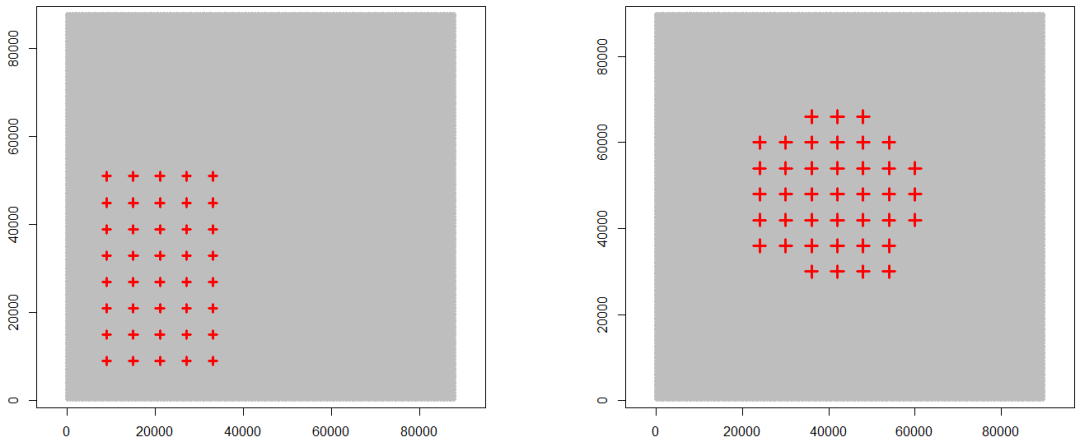

Figure 2.4: An example of the regular grid used for homogeneous density (left) and one realisation of a random regular grid for inhomogeneous density (right) superimposed on the mask. The red crosses indicate traps and the axes show coordinates in metres.

2.6 Simulation-Based Evaluation

A simulation study was conducted using the secrdesign R package (Efford, 2020) to evaluate the performance of the proposed optimal design configurations. The study included a total of 36 design configurations for each case study, covering various scenarios: three design approaches (regular grid, min(n,r), and $Q_{\bar{p}_m}$), large and small study areas, detector counts of 20 or 40, and different density surfaces (homogeneous, gradient, and conical).

The initial goal was to gain a sense of the quality of the data before fitting SCR models and evaluating the resulting metrics. For each design configuration, summary statistics such as $E(n)$, $E(r)$, and $E(m)$ were calculated to ensure that the expected number of captures, recaptures and movements (m) were realistic. Movements are a subset of recaptures, and occur when an individual is recaptured at a different site from its previous capture location (Efford & Boulanger, 2019). Then, using a given trap design and case study specific parameter values, individual activity centres and capture histories were simulated, assuming count proximity detectors and a Poisson distribution for the activity centres and capture frequencies. The observed outcomes were then compared with what was expected to assess the quality of the data.

Next, SCR models could be fitted using simulated data, which produced estimates of density, amongst other model parameters. When fitting models for the inhomogeneous density surfaces, the correct density model was specified; this was optional. When using homogeneous density, a large study area and 40 detectors for one case study, three average density estimates with $n = 100$ were produced. These were for the regular grid, the min(n,r) designs, and the clustered $Q_{\bar{p}_m}$ designs. These density estimates were assessed in terms of bias, using percent relative bias (RB), and precision, using percent estimated relative standard error (RSE), whose respective formulae can be seen in Equations 2.6 and 2.7, below. The approximation of CV($\widehat{D}$) from Equation 2.3 was also calculated for all design configurations and compared to the empirical estimate of CV($\widehat{D}$), the RSE, produced via simulation.

$$RB = \left(\frac{\hat{D} - D}{D} \right) * 100 \tag{2.6}$$

$$RSE = \left(\frac{SE(\hat{D})}{\hat{D}} \right) * 100 \tag{2.7}$$

3 Results

3.1 Designs

First, a post hoc assessment of the designs is done to investigate patterns and differences across the configurations and case studies. One realisation of a design for each type of density pattern, over a large study area and 40 detectors, can be seen in Figures 3.1, 3.2 and 3.3. These are the min(n,r) and $Q_{\bar{p}_m}$ designs for the leopard, deer and rabbit respectively. Across all case studies, it is evident that the min(n,r) criterion produced more spread-out designs compared to the clustered designs produced by the $Q_{\bar{p}_m}$ criterion, which is to be expected.

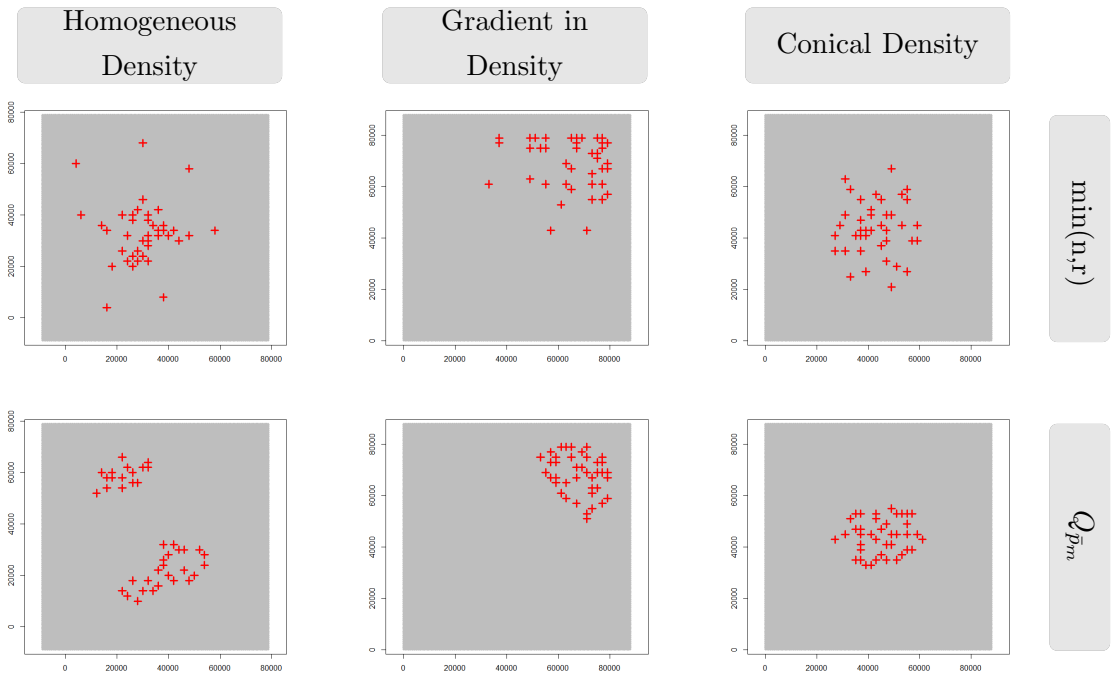

Figure 3.1: Examples of designs generated using the optimisation algorithm with the min(n,r) and $Q_{\bar{p}_m}$ criteria for a large area with 40 traps for the leopard across the three density surfaces. The red crosses indicate detectors.

When looking at homogeneous density for the leopard, it is clear that the min(n,r) criterion attempted to balance n and r since it has a sufficient number of traps closer together to gain meaningful recaptures, as well as adequate traps far out to capture new individuals (Figure 3.1). This reflects a balance between a space-filling and clustering design. On the other hand, the $Q_{\bar{p}_m}$ criterion has only formed clusters which seek to increase the number of recaptures. The designs for the inhomogeneous densities follow a similar pattern, bar that the traps are solely concentrated in the high density areas, the top right corner (gradient) and the centre (conical).

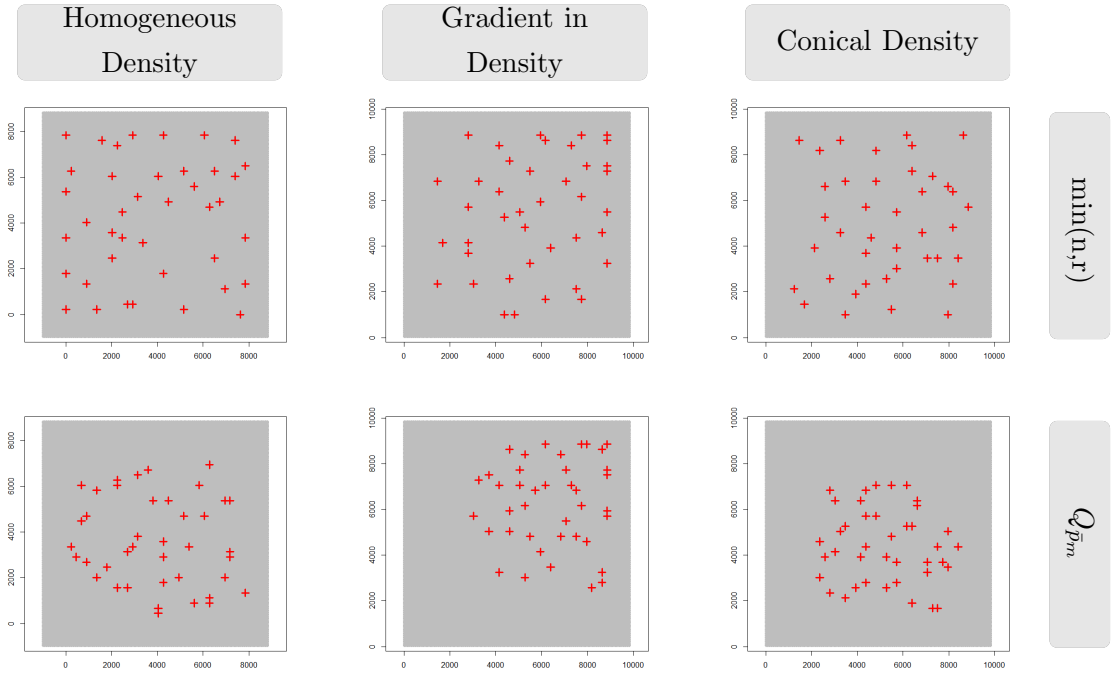

Figure 3.2: Examples of designs generated using the optimisation algorithm with the min(n,r) and $Q_{\bar{p}_m}$ criteria for a large area with 40 traps for the deer across the three density surfaces. The red crosses indicate detectors.

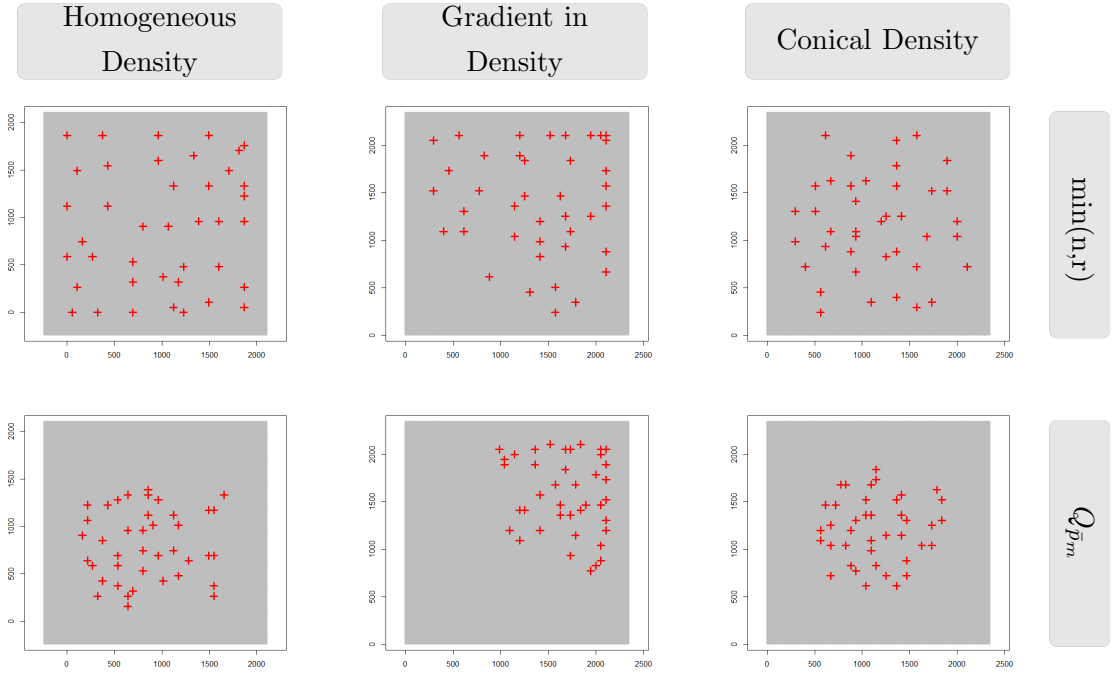

Figure 3.3: Examples of designs generated using the optimisation algorithm with the min(n,r) and $Q_{\bar{p}_m}$ criteria for a large area with 40 traps for the rabbit across the three density surfaces. The red crosses indicate detectors.

The designs for the deer and the rabbit exhibit quite similar patterns to each other, despite their substantially different parameters. For all density patterns, the min(n,r) criterion dispersed the traps very widely, covering almost the entire study area, reflecting a more space-filling design which did not occur with the leopard. In the case of the $Q_{\bar{p}_m}$ criterion, the traps are less spread out, but the clustering quality seen for the leopard designs is not as present for the deer and rabbit case studies. Therefore the inhomogeneous density patterns placed the traps in order to sample not only from the highest density areas but also the less dense areas, particularly for min(n,r) designs (Figures 3.2 and 3.3).

Although only the large study area with 40 detectors is displayed, the other configurations with the small study area or 20 detectors followed similar patterns to those discussed.

3.2 Simulated Data Quality

In addition, the quality of the simulated datasets is assessed by comparing expected detection data to observed data via simulation before fitting SCR models. It is essential to understand whether the proposed designs and parameter values produce datasets with reasonable captures, recaptures, and movements compared to the expected values before fitting SCR models.

When comparing the expected captures $E(n)$, recaptures $E(r)$, and movements $E(m)$ to what was produced across all configurations, it is clear that the simulated datasets produced almost exactly what was expected for n and r, with slight variations in some cases. All results are summarised in Appendix A, Tables A.1, A.2, A.3 for the leopard, deer and rabbit, respectively. Due to the nature of the detector type, an individual may be detected more than once on one occasion, where the detection time is recorded but not used in the model within each occasion (the norm in SCR). Therefore, some movements may go unrecorded, leading to an underestimation of the animal's activity. Without the precise timing information, the observed number of movements was always lower than expected.

In all homogeneous density regular grid scenarios, one regular grid was scaled by σ for each species, meaning all expected data is a singular value. Whereas for both inhomogeneous density surfaces, there were 100 random variations of the regular grid. In these cases, the regular grid was not optimised to be placed in high density areas. Therefore, there is a large variation in the data for the regular grids, as seen in Figures 3.4, 3.5 and 3.6. These figures show the expected and observed detection data of the large study area with 40 traps, representing the patterns seen across most

combinations of area size and number of traps for each density surface. Further note that whilst the absolute numbers differ between scenarios, all configurations tend to follow the same trends and patterns.

One noticeable trend across all case studies and density surfaces is that the configurations with 20 traps have lower total counts than those with 40 traps, approximately half. This is expected since more traps allow for more detections. However, Tables A.2 and A.3 in Appendix A show that the deer produce the same detection data when using 20 or 40 traps in the small study area of the homogeneous density surface. This is also true for the rabbit. This could indicate some form of upper bound to n, r and m when there is homogeneous density over a small area for case studies with similar parameters to the deer and rabbit, and therefore detection data in a small area is independent of the number of traps used.

In the case of the leopard, it is evident in Figure 3.4 that all density surface types for the min(n,r) designs result in an equal number of captures and recaptures, where the number of recaptures excludes the first captures. However, there tend to be more captures than recaptures for the regular grid.

Furthermore, over all the density surfaces for the leopard, the $Q_{\bar{p}_m}$ designs yield the highest recaptures within their respective density surface. This is due to the clustered nature of the $Q_{\bar{p}_m}$ designs as previously seen in Figure 3.1. However, the conical density surface has the highest n, r and m values across the three density surfaces, with the homogeneous density surface having the lowest, as seen in Figure 3.4. The clustered designs from Figure 3.1 benefit the inhomogeneous density surfaces as the algorithm can sample from the high density areas, therefore the homogeneous density surface has the lowest absolute values of the three surfaces.

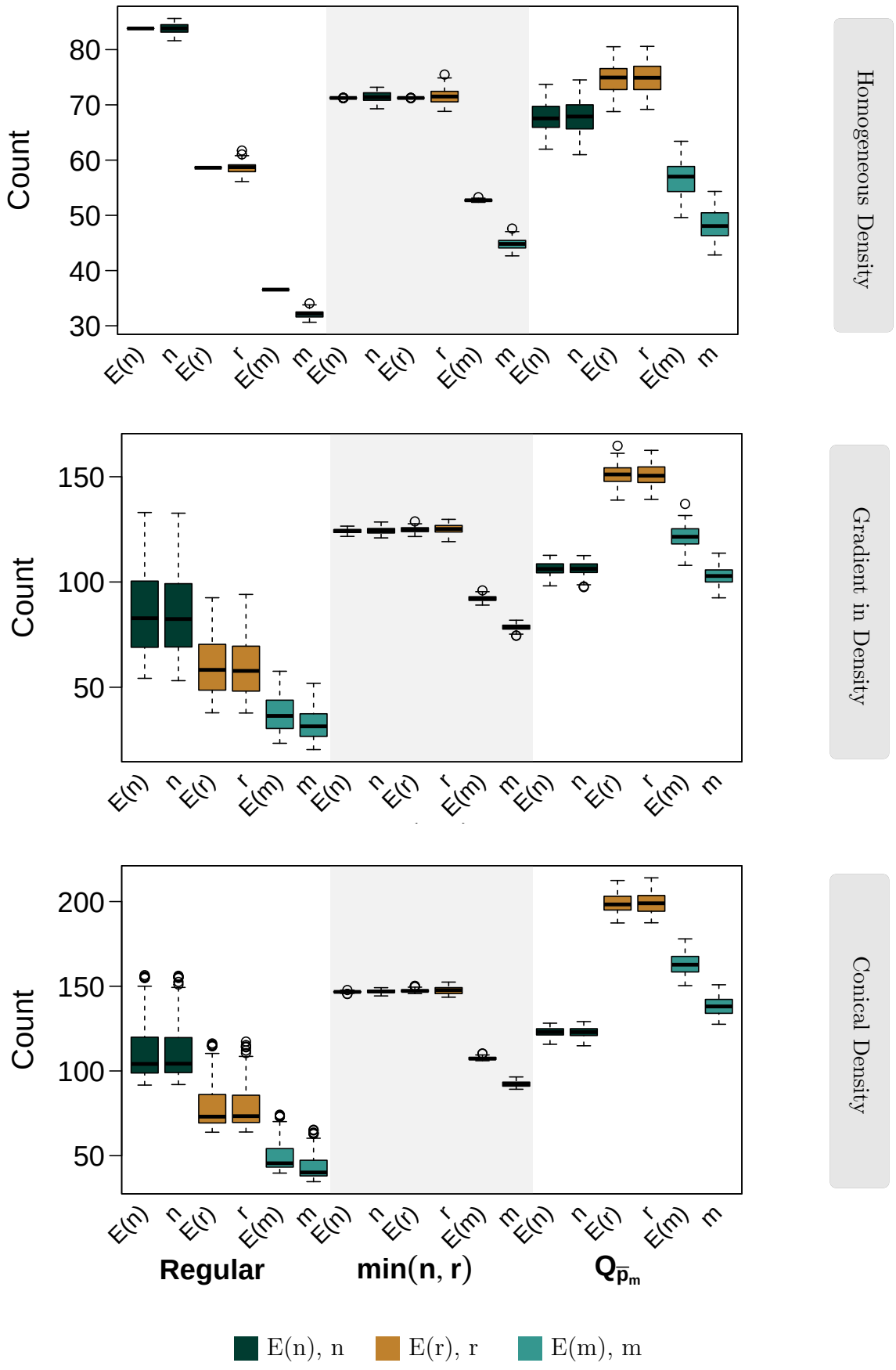

Figure 3.4: Boxplots of the expected and observed detection data for the leopard for 100 realisations of each design; regular, min(n,r) and $Q_{\bar{p}_m}$, prior to simulation for a large area with 40 detectors across the different density surfaces; homogeneous density (top), gradient in density (middle) and conical density (bottom).

In contrast to the leopard, the deer and rabbit exhibit similar patterns across their configurations (Figures 3.5 and 3.6), showing more recaptures than captures. This implies that these case studies, on average, recapture an individual more than once. This may be due to the more balanced designs informed by λ_0 values, seen in Figures 3.2 and 3.3, where traps are far less clustered than in the leopard case study. Across all density surfaces for the deer and rabbit, the min(n,r) designs produce the highest number of captures. In the inhomogeneous density surfaces, the $Q_{\bar{p}_m}$ designs produce the highest number of recaptures. In contrast, the regular grid produces the highest number of recaptures across the homogeneous density surfaces.

The deer produced exceedingly high recaptures relative to the other two case studies. This may be due to the deer's high detectability. However, there is a major discrepancy between the expected and observed movements of the deer. This could be due to incomplete observations of the temporal sequence of the numerous r detections.

While certain trends are apparent, notable exceptions exist, such as the deer and rabbit conical density surfaces for the small area with 40 traps, seen in Appendix A, Figure A.1. The primary distinction is that the regular grid for the conical density exhibits significantly less variability than other configurations. This is attributed to covering an extensive portion of the area with high density.

Overall, the detection data for the deer is much higher than for the leopard and rabbit. The deer is a high density species, so it is expected that it is more easily observed.

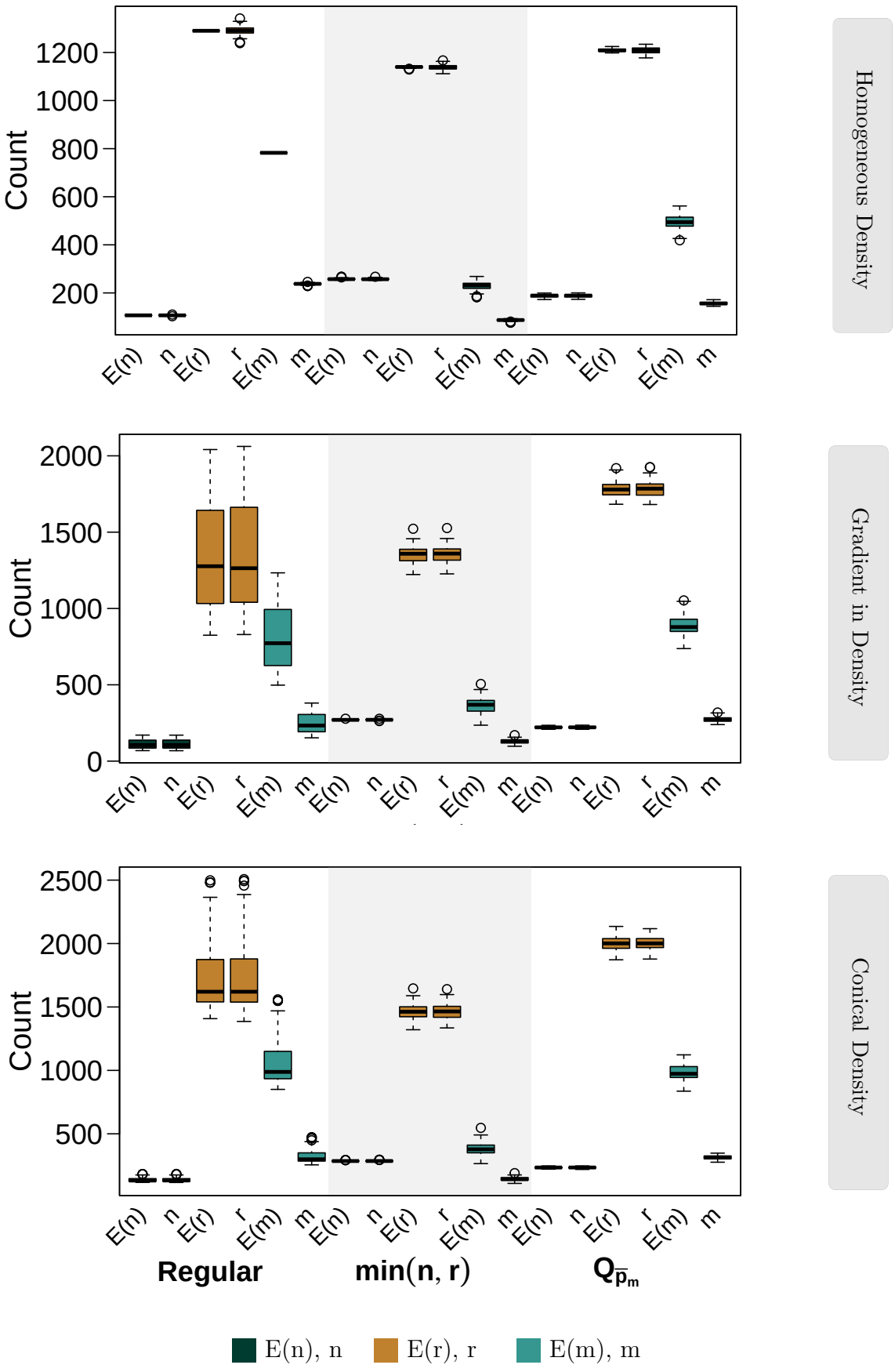

Figure 3.5: Boxplots of the expected and observed detection data for the deer for 100 realisations of each design; regular, min(n,r) and $Q_{\bar{p}_m}$, prior to simulation for a large area with 40 detectors across the different density surfaces; homogeneous density (top), gradient in density (middle) and conical density (bottom).

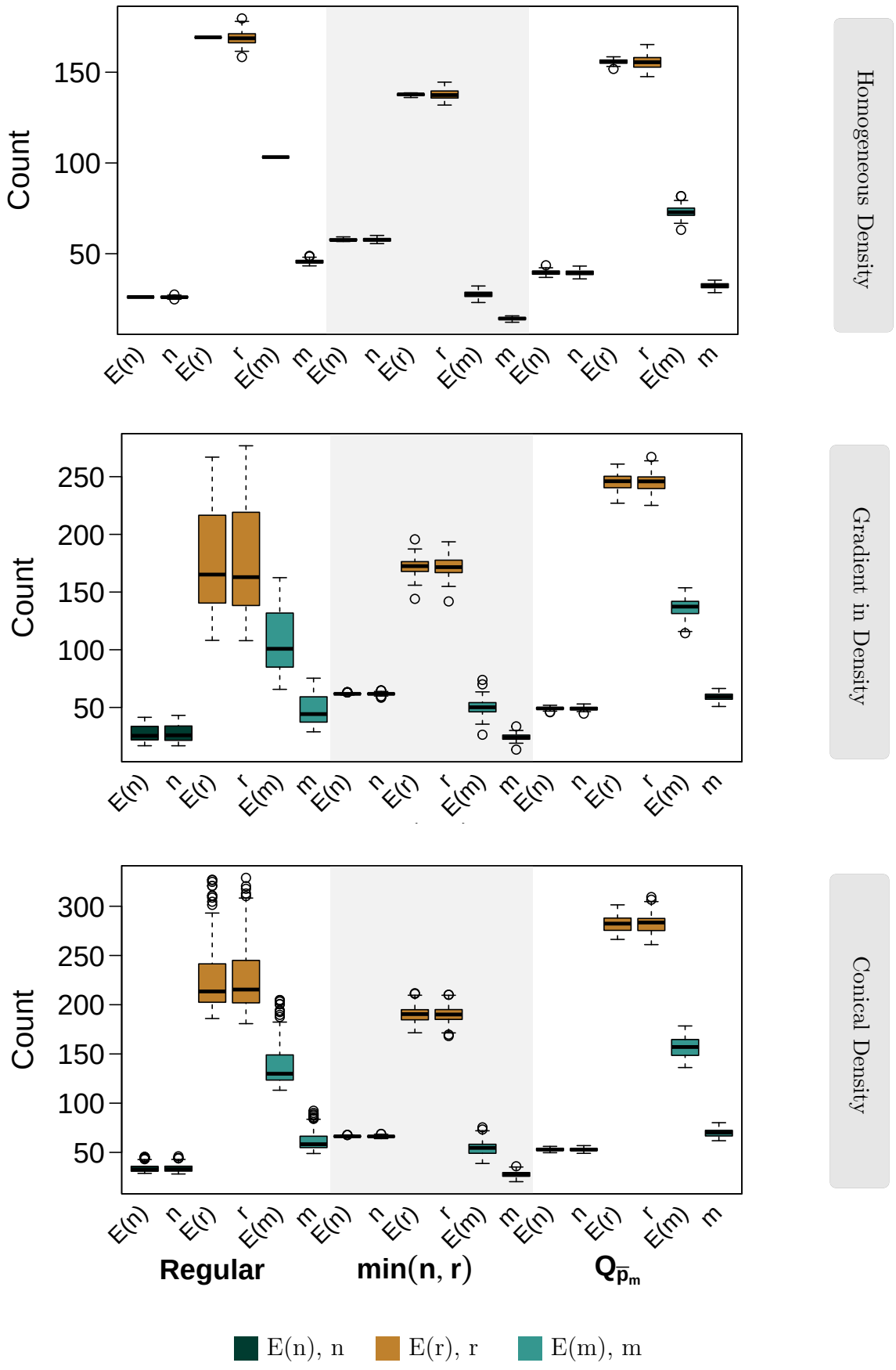

Figure 3.6: Boxplots of the expected and observed detection data for the rabbit for 100 realisations of each design; regular, min(n,r) and $Q_{\bar{p}_m}$, prior to simulation for a large area with 40 detectors across the different density surfaces; homogeneous density (top), gradient in density (middle) and conical density (bottom).

3.3 Evaluation of Density Estimates

The initial focus in evaluating the results from the fitted models in the simulation study is on the relative bias of the density estimates. SCR estimators are generally known to be unbiased for a variety of detector arrangements; however, a post hoc assessment of possible bias is always recommended, especially in the case of inhomogeneous density. Additionally, it is crucial to investigate the relative standard error of the density estimates as the design optimisation focused on maximising approximate precision, assessed through RSE. The min(n,r) criterion attempted to maximise the precision of the density estimators or, equivalently, minimise CV($\widehat{D}$). Thus, the CV($\widehat{D}$) approximation for all design configurations was also calculated to be compared to the observed RSE. An in-depth summary of all these results can be seen in Appendix B in Tables B.1, B.2 and B.3.

3.3.1 Homogeneous Density

Results from the homogeneous simulations across all case studies are encouraging, showing relatively low bias, in keeping with the allowable bias of ±5%, for most configurations, as highlighted by the left hand side plots in Figure 3.7. The regular grid, min(n,r), and $Q_{\bar{p}_m}$ designs all produced unbiased estimates for the leopard and the deer when using 40 traps in either study area. The RSE is 15% or less for these configurations, which is below the acceptable threshold of 20%. Although, the variation is consistently lower for the deer across all designs compared to the leopard, as seen in the right hand side plots of Figure 3.7, which can be attributed to the much higher sample size of the deer.

When the number of detectors was decreased to 20, a slight bias and higher variation was introduced in the leopard's large and small study areas, with the density being overestimated, and the exception of the min(n,r) design in the large study area. With insufficient traps, min(n,r) designs performed better in the large study area, whereas $Q_{\bar{p}_m}$ designs performed better in the small study area. Yet it is clear that in each case, both design approaches produced results as good as or better than the regular grid, which was not unbiased with 20 traps. On the other hand, the designs generated for the deer with only 20 traps result in unbiased, low variance estimates across all configurations (Figure 3.7).

However, the designs for the rabbit exhibit substantially differing results between the configurations, as seen in the third row of Figure 3.7. From these, it is clear that the simulations produced highly variable estimates compared to the other two case studies. The $Q_{\bar{p}_m}$ designs did not produce unbiased estimates; however, they

performed the most consistently across rabbit configurations, with the RB still being within ±5% and a relatively low RSE. Only the regular grid and min(n,r) designs give rise to unbiased estimates, although these must be considered in tandem with the high RSEs.

Specifically, looking at the min(n,r) designs for the rabbit configuration of 20 traps in a large area, there was evidence of insufficient data quality and, thus, poor model performance. This can be seen by the poor density estimates in the left boxplot in Appendix B, Figure B.1. Extreme density estimates and estimates with infinite or severe standard errors were removed, as they indicated that there had been a convergence issue. Therefore, approximately one-third of the estimates were removed. A boxplot showing the distribution of the density estimates after weak data was removed can be seen in comparison in the right boxplot in Figure B.1. Therefore, only 67% of the designs produced strong enough data to fit a model, which then produced the unbiased estimates for this configuration (Figure 3.7).

Lastly, the approximate CV($\widehat{D}$) results, summarised in the Appendix, Tables B.1, B.2 and B.3, are largely in keeping with those produced by Efford and Boulanger (2019). The approximation almost always underestimates RSE, but is more accurate when RSE $< 20\%$. This implies that when the optimisation algorithm was maximising precision using the approximation of CV($\widehat{D}$), it overestimates how well the design will actually perform.

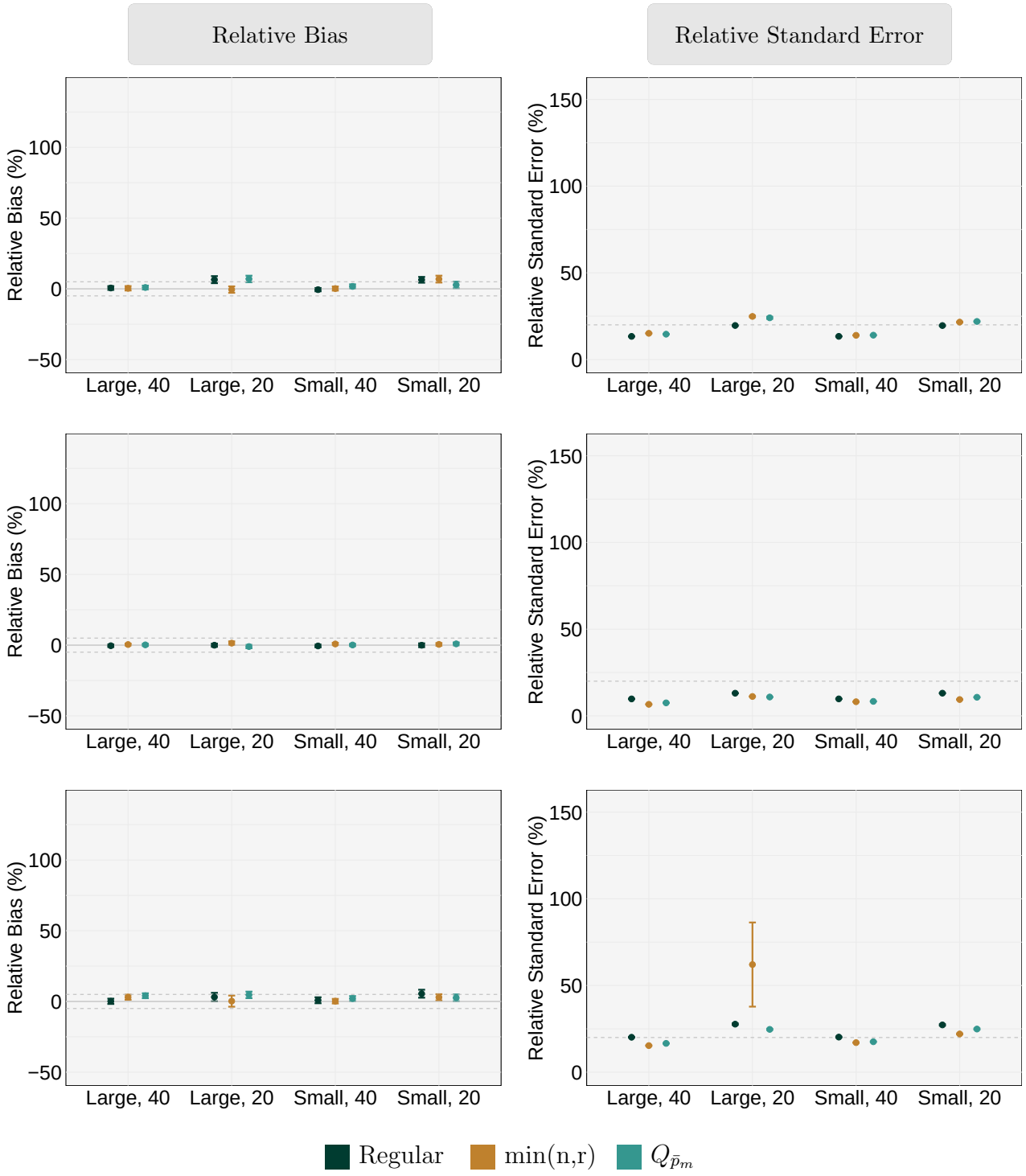

Figure 3.7: Average percent RB (left hand side) and percent RSE (right hand side) of homogeneous density estimates for the leopard (top row), deer (middle row) and rabbit (bottom row). Vertical lines are standard error bars, most of which are too small to be seen. The three designs, regular, min(n,r) and $Q_{\bar{p}_m}$ are represented by the three colours as indicated by the key. The horizontal dashed grey lines on the RB plots represent an allowable amount of bias (±5%), with the solid grey line representing no bias in the estimates. The horizontal grey dashed line on the RSE plots indicates an acceptable level of RSE (20%).

3.3.2 Inhomogeneous Density

When spatially varying density models were used, the density estimates across all configurations are seen to be far more biased and variable than when assuming homogeneous density, as is summarised in Figures 3.8 and 3.9 when compared to Figure 3.7. A post-hoc assessment of the density estimates across scenarios showed that, once again, a handful of simulated populations had produced infinite standard errors, indicating an issue with model convergence. These were therefore removed before the calculation of summary statistics. However, the configurations that exhibited high bias did not have the extreme or outlying estimates removed since they were not so severe as to suggest an issue but rather were indicative of poorly performing trap designs.

The most noticeable observation across both density patterns and proposed optimal designs is that in most cases, the density is severely overestimated for the leopard and underestimated for the deer and rabbit. Although the estimates were not expected to show such a high level of bias, this aligns with the design characteristics previously discussed (Figures 3.1, 3.2 and 3.3). The designs generated for the leopard concentrated traps only in the high density areas as this is where the optimisation algorithm recognised it was capturing the best data. On the other hand, the designs outputted for the deer and the rabbit placed traps far more widely, meaning they would be sampling very low density areas (Figures 3.2 and 3.3).

Although the results for the leopard case study are biased, they show low variability, as indicated by the top row of plots in Figures 3.8 and 3.9. Even in certain cases with very high bias, such as the small study area with 20 traps, the density estimates are consistently higher, indicating a shift in the entire distribution rather than increased variability. This is further illustrated by the boxplots of these distributions in Appendix B, Figure B.2.

Additionally, for the leopard, the small study area performed better than the large area, especially with 40 traps, as was to be expected, as the area is adequately surveyed. The min(n,r) designs were also slightly less biased than the $Q_{\bar{p}_m}$ designs across all configurations. Still, both design approaches produced considerably worse results than the standard regular grid. Furthermore, the gradient in density scenarios outperformed the conical density scenarios, which was not expected as the nature of the gradient does not allow for the highest density area to be completely sampled.

In contrast, the gradient in density scenarios for the deer and rabbit performed more poorly than the conical density. This was to be expected as traps were spread far

apart, as seen in Figures 3.2 and 3.3, and were not always able to sample from high density areas. The min(n,r) and $Q_{\bar{p}_m}$ designs performed better in terms of bias with only 20 traps as the algorithm was underestimating to a lesser extent, however, these estimates were generally more variable than those with 40 traps. This can be seen in the second and third row of Figures 3.8 and 3.9.

On the other hand, the regular grid for the conical density overestimates the density of both deer and rabbit. This is attributed to the random generation of the regular grid, as discussed earlier, and its inconsistent coverage of high density areas. The optimisation approaches incorporate the varying density surface when generating designs, whereas the regular grid does not. As a result, the regular grid is at a disadvantage and has considerable variability in the inhomogeneous density surfaces. In the small area with 40 traps, this pattern persists but with slightly less variation due to the wider distribution of traps, which provides more comprehensive coverage of the entire area.

Again, the trends in approximate CV($\widehat{D}$) are similar to those seen by Durbach et al. (2021), despite the results in this paper being far more biased. The margin by which the approximation underestimates RSE is larger under inhomogeneous density than homogeneous density, and this effect is more severe when RSE $> 20\%$. This is especially noticeable in the case of the rabbit.

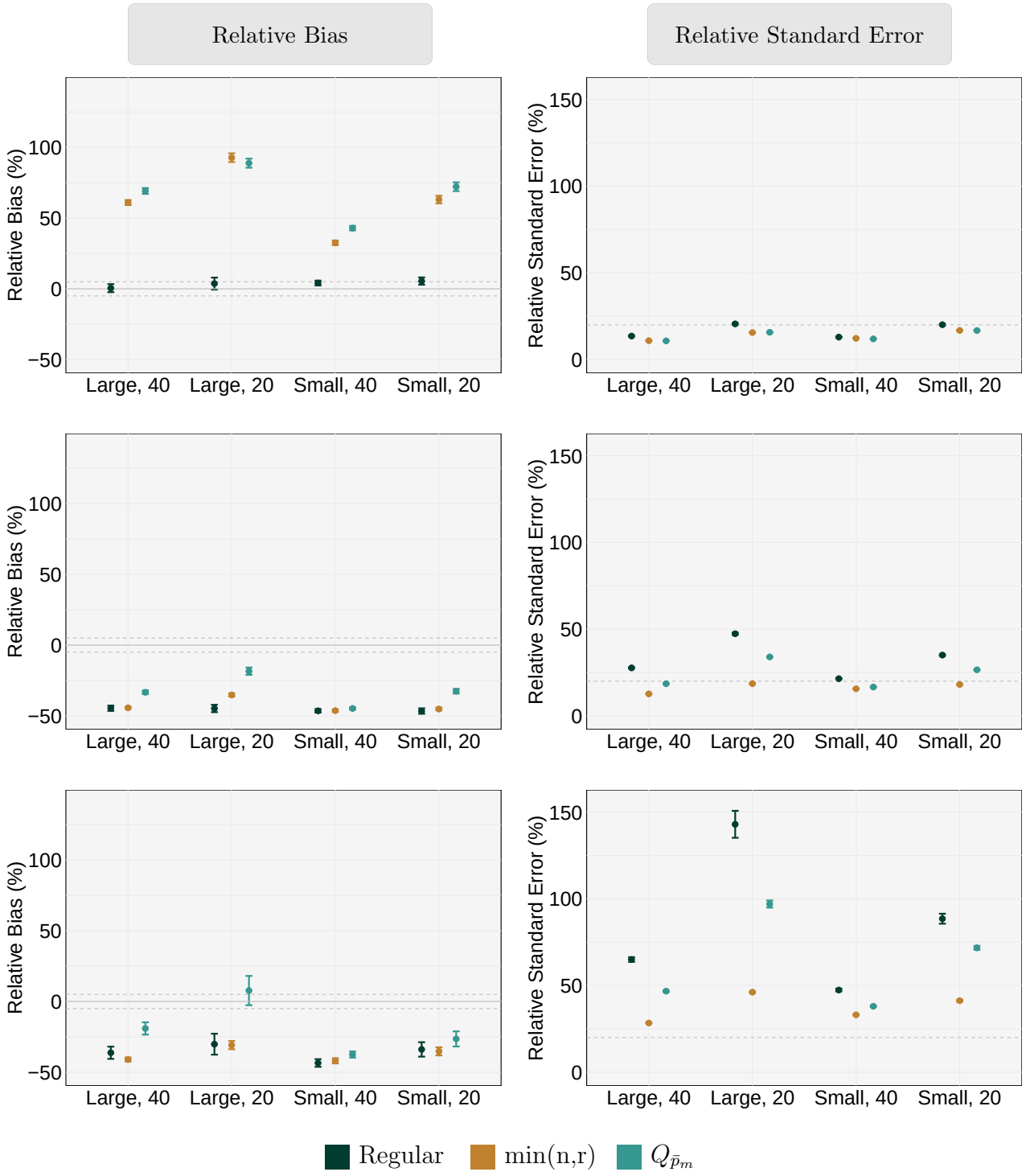

Figure 3.8: Percent RB (left hand side) and percent RSE (right hand side) of the gradient in density estimates for the leopard (top row), deer (middle row) and rabbit (bottom row). Vertical lines are standard error bars, most of which are too small to be seen. The three designs, regular, min(n,r) and $Q_{\bar{p}_m}$ are represented by the three colours as indicated by the key. The horizontal dashed grey lines on the RB plots represent an allowable amount of bias (±5%), with the solid grey line representing no bias in the estimates. The horizontal grey dashed line on the RSE plots indicates an acceptable level of RSE (20%).

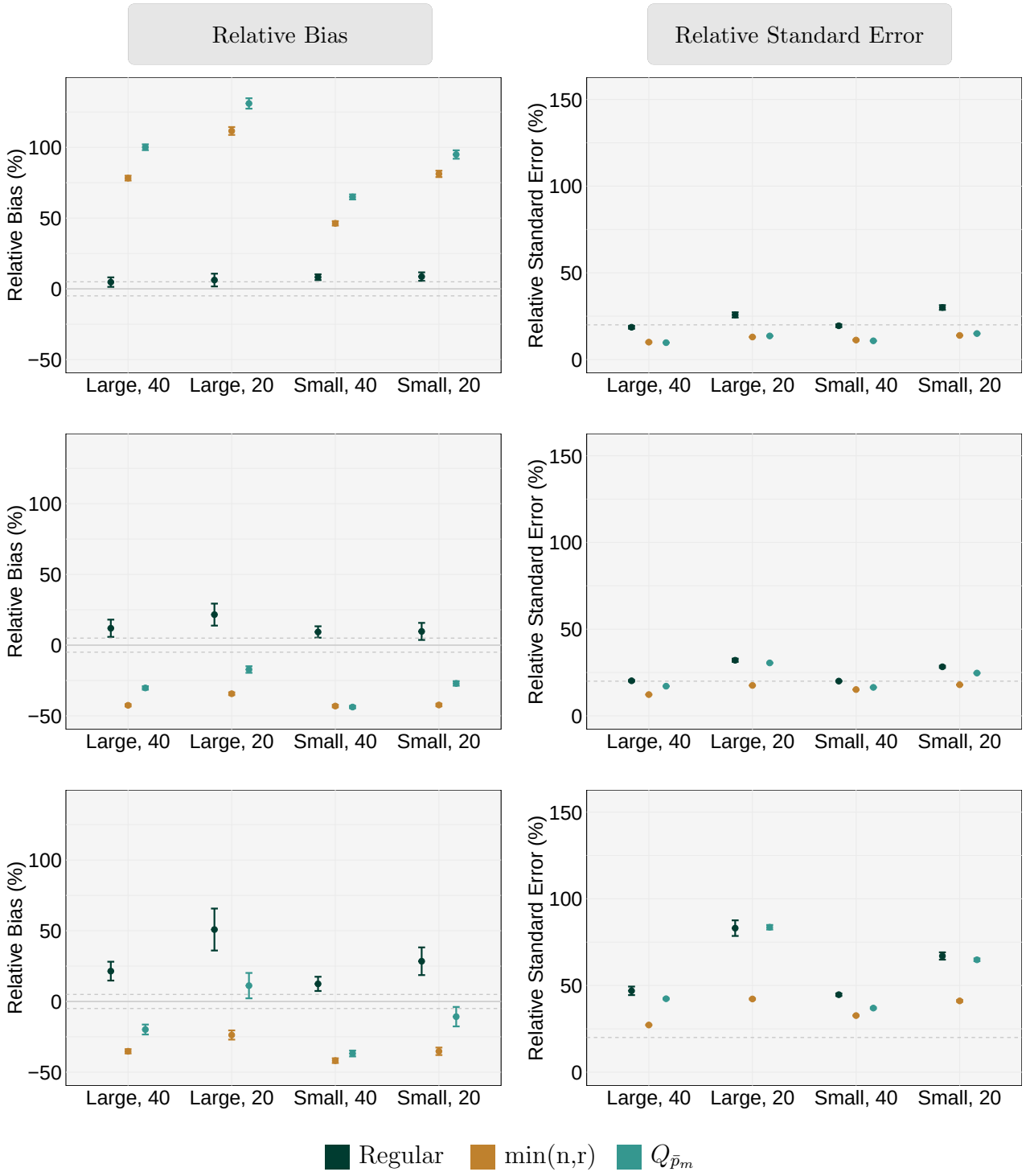

Figure 3.9: Percent RB (left hand side) and percent RSE (right hand side) of conical density estimates for the leopard (top row), deer (middle row) and rabbit (bottom row). Vertical lines are standard error bars, most of which are too small to be seen. The three designs, regular, min(n,r) and $Q_{\bar{p}_m}$ are represented by the three colours as indicated by the key. The horizontal dashed grey lines on the RB plots represent an allowable amount of bias (±5%), with the solid grey line representing no bias in the estimates. The horizontal grey dashed line on the RSE plots indicates an acceptable level of RSE (20%).

4 Discussion

The primary aim of this study was to evaluate the effectiveness of various design optimisation methodologies in estimating density using SCR models for a range of case studies. The findings indicate that when density is homogeneous, the optimisation approaches mostly produce designs that are competitive with the regular grid across case studies. However, when density varies spatially, the accuracy and precision of the estimates are heavily affected, with the specific parameters of the species influencing the nature of the designs generated. Furthermore, the results suggest that perhaps 40 detectors are insufficient to adequately sample the range of spatial variation in density, affecting the accuracy of the density estimates. These findings contribute to the existing body of knowledge by providing a framework for optimising trap designs suited to specific ecological characteristics of different species, such as leopards, roe deer, and Sumatran striped rabbits.

The data quality assessment results before model fitting appeared promising, with the observed detection data closely matching what was expected. However, this did not necessarily always translate into accurate estimates of density. This can be potentially attributed to the optimisation algorithm not assessing the recaptures' usefulness but only aiming to increase the absolute number. Thus, whilst the simulations may have produced data near the expected detection data, it was not always sufficiently informative to accurately estimate density.

Additionally, this study found that the magnitude of the encounter rate at a distance of zero, λ_0, affects the number of captures versus recaptures. When λ_0 is low, such as in the case of the leopard, captures and recaptures tend to be equal due to the elusiveness of the species. On the other hand, the more detectable species, such as the rabbit and especially the deer, exhibit a higher number of recaptures, as seen in Figure 3.5.

In light of this, it was further found that the optimisation algorithm was affected by the relative size of λ_0. For a low λ_0, such as for the leopard, the algorithms struggled to increase the number of recaptures and thus placed traps closer together as seen in Figure 3.1. Consequently, the min(n,r) designs, which attempt to balance n and r, generally performed better with a lower λ_0 than the clustered $Q_{\bar{p}_m}$ designs, which only aim to maximise r. On the contrary, species with a higher λ_0, such as the deer, resulted in the min(n,r) criterion generating far more spread out designs, as seen in Figure 3.2. In these cases, min(n,r) designs did not perform as well as the regular grid or even $Q_{\bar{p}_m}$ designs, specifically when density varied spatially. This is because

the designs are unable to produce meaningful spatial recaptures as the algorithm attempts to spread traps out over a large area.

In general, the differences in performance between the regular grid and proposed optimal designs are particularly pronounced when comparing scenarios with homogeneous versus inhomogeneous density. When assuming homogeneous density, the min(n,r) and $Q_{\bar{p}_m}$ designs are competitive with the regular grid for the most part, in some cases even outperforming the regular grid when provided with sufficient traps. Conversely, under inhomogeneous density, estimates varied widely and were severely impacted by the nature of the designs produced by the optimisation algorithm.

As noted by Dupont et al. (2021), sampling only areas with high densities of individuals leads to overestimates of density, while sampling away from these areas does the opposite. When density varied spatially, this was attributed to the model incorporating data from nearby traps in an inherently spatially autocorrelated landscape (Dupont et al., 2021), thereby somewhat explaining the poor performance seen in this paper under inhomogeneous density. Furthermore, Durbach et al. (2021) highlight that adequate coverage of covariate space is essential because unbiased density estimates rely on accurately estimating covariate relationships, which is more achievable when the covariate space has been thoroughly sampled. The same can be said for modelling spatially varying density.

From a case study-specific perspective, the most elusive and wide-ranging species, the leopard, demonstrates dissimilar results from the deer and rabbit. Under homogeneous density with 40 traps, the proposed optimal designs are competitive with the regular grid. Interestingly, with only 20 traps, the min(n,r) design outperforms the regular grid in the large study area, while the $Q_{\bar{p}_m}$ design achieved the most unbiased results in the small study area. Nevertheless, the slight bias and increased RSE of those configurations with 20 traps highlight that 20 detectors are inefficient in monitoring this species, and this effect is further exacerbated under inhomogeneous density.

As seen previously, modelling with spatially varying density resulted in extreme, albeit precise, overestimates of leopard density due to the clustering of traps in high density areas. Whilst results from Durbach et al. (2021) were unbiased when estimating spatially varying density for leopards, the high density areas were more dispersed and irregular, meaning that their spatial coverage was smaller in proportion to the total landscape. This possibly led to swift saturation of the traps in high density regions, as fewer traps were needed to cover these areas fully. Additionally, unlike in this paper, the few traps placed in low-density areas may

have been sufficient for capturing a representative sample. Therefore, the proposed optimal designs cannot be considered competitive with the regular grid when density varies spatially, as the regular grid produced nearly unbiased estimates and would likely perform even better if strategically placed in the areas of the highest density. Nevertheless, it is possible that with more detectors, the proposed optimal designs would perform just as well, if not better, than the regular grid because there would be enough traps to spread out over the full study area and thus also sample the lower density areas.

In contrast, assuming homogeneous density for species like roe deer, which have relatively high density and detectability, allows for accurate surveys regardless of the study area's size or number of traps. When using a sufficient number of traps (40) or a smaller study area, both the min(n,r) and $Q_{\bar{p}_m}$ designs slightly outperform the regular grid, producing unbiased and precise estimates. Despite this, these designs should be seen as flexible alternatives when a regular grid is not feasible rather than universally optimal solutions.

The third case study was the most unique due to the Sumatran striped rabbit's elusive nature, small movement range and relatively low density. Consequently, it had the most variable and unexpected results. Under homogeneous density, only the regular grid produced consistently reliable results across large and small study areas and only for 40 traps. Specifically, the proposed optimal designs can only be seen as competitive with the regular grid in a small study area with sufficient traps. Min(n,r) designs were unbiased and acceptably precise, whilst $Q_{\bar{p}_m}$ designs were only slightly biased and still precise. Given the rabbit's λ_0 value and respective dispersed trap designs, this result is not surprising.

Across all configurations, 20 traps proved insufficient in surveying the rabbit, with the caveat of the min(n,r) designs in the large study area. While this configuration yielded unbiased estimates, it is highly variable and requires approximately one-third of the data to be removed. This configuration does not consistently provide strong data for the SCR model to produce accurate estimates. However, when sufficient data is available, the estimates are reliable. This underscores the necessity for additional traps when surveying a species with characteristics similar to the rabbit case study to mitigate the risk of inadequate data capture, especially considering the significant time and resource investments involved.

When looking at inhomogeneous density for the deer and the rabbit, the performance of the proposed optimal designs decline, leading to underestimates of density in both

density-varying scenarios. This can be attributed to a high λ_0 value and thus the wide dispersion of traps previously discussed, with $Q_{\bar{p}_m}$ designs performing slightly better than min(n,r) designs in terms of bias but not precision. Unexpectedly, these scenarios with only 20 traps performed better than those with 40. This is possibly because fewer traps would cause the GA not to spread the traps as far to ensure sufficient recaptures and thus be more focused in the high density areas. Furthermore, the designs produced for conical density performed better than those for the gradient in density, as was to be expected because the gradient in density surface has the highest density areas in the buffer region of the mask. This means that the traps could not adequately sample the highest density area, particularly given the spread-out nature of the designs.

The regular grid design exhibited poor performance across all inhomogeneous configurations, particularly the gradient in density, largely due to its random generation preventing detectors from being centred on areas of high density. This disadvantaged it compared to other algorithms, which had access to information about the high-density regions. Given this, it cannot be said with certainty that the optimisation approaches are competitive with the regular grid when density varies spatially. If more traps had been used or the regular grid had been optimised to target high-density areas, performance across all configurations might have improved, suggesting a valuable opportunity for future research.

A further potential limitation to consider is the possibility of a convergence issue with the GA. Although this is highly unlikely as it generally converges within 50 iterations (Durbach et al., 2021), there is a chance that more iterations may have been necessary to reach the optimal solution in some cases. This raises a possible area for future investigation, where running the GA for more generations could potentially yield better results. Moreover, using additional replicates or iterations of the designs and simulations may have enhanced the precision and consistency of the findings.

5 Conclusion

This paper evaluates the performance of various trap design optimisation methods across three species with distinct ecological characteristics. It provides a framework that ecologists can use to make informed decisions about survey designs. The findings highlight the importance of considering species-specific movement patterns and detectability when planning camera trap surveys, as these factors significantly influence the effectiveness of different design approaches.

This study highlights the need for more traps and a strategic placement that balances the number of captures and recaptures for species with low detectability and wide-ranging movements, such as leopards. The min(n,r) designs showed promise in this type of scenario, suggesting that focusing on the number of detections and the spatial distribution of recaptures is crucial. Conversely, for a more detectable species like the roe deer, the results indicated that a regular grid or designs that maximise recaptures, such as $Q_{\bar{p}_m}$, can be effective, even with a smaller number of traps or in larger study areas.

The Sumatran striped rabbit presented a challenging case with its unique combination of low density, high elusivity, and small movement range. The findings suggest that a regular grid may be the most reliable approach for such species, but only when sufficient traps are used.

Moreover, the results indicated that, under homogeneous density conditions, optimised designs were competitive with regular grids. However, when density varied spatially, performance was more variable and heavily influenced by species-specific characteristics, particularly detectability. This significantly impacted the quality of detection data, as weak designs failed to adequately capture the range of variation across the area. These findings are crucial for real-world surveys, where animal populations are often unevenly distributed, suggesting that regular grids may be preferable when a density covariate is assumed.

This research contributes to existing knowledge by providing a focused understanding of how different optimisation criteria perform across various ecological contexts. It challenges the reliance on broad, heuristic guidelines for trap placement and advocates for a more customised approach that considers the specific characteristics of the studied species. Additionally, it emphasises the importance of simulation studies before fieldwork, where inefficient survey designs may squander already limited resources. It underscores the need for precision in wildlife monitoring to support conservation and management efforts.

A Simulated Data Quality

Table A.1: Table of the average expected and observed detection data for the leopard for all designs; regular, min(n,r) and $Q_{\bar{p}_m}$ and the various density surfaces; homogeneous density, gradient in density and conical density. $E(n)$ and n are the expected number of captures and observed captures respectively. $E(r)$ and r are the expected number of recaptures and observed recaptures respectively. $E(m)$ and m are the expected number of movements and observed movements respectively. All values were rounded to the nearest whole number.

	Regular						*min(n, r)*						$Q_{\bar{p}_m}$					
	E(n)	**n**	**E(r)**	**r**	**E(m)**	**m**	**E(n)**	**n**	**E(r)**	**r**	**E(m)**	**m**	**E(n)**	**n**	**E(r)**	**r**	**E(m)**	**m**
Homogeneous Density																		
Large Area																		
40	84	84	59	59	37	32	71	71	71	72	53	45	68	68	75	75	57	48
20	43	43	28	28	17	15	36	36	36	36	26	22	33	33	38	38	30	25
Small Area																		
40	84	84	59	59	37	32	71	71	71	71	52	45	62	62	81	81	63	54
20	43	43	28	28	17	15	36	36	36	36	26	22	32	32	40	40	31	26
Gradient in Density																		
Large Area																		
40	87	87	61	60	38	33	124	124	125	125	92	79	106	106	151	151	122	103
20	46	46	29	29	17	15	70	70	71	70	52	44	61	61	79	78	62	52
Small Area																		
40	87	87	61	61	38	33	99	99	100	100	74	63	85	85	134	134	110	93
20	46	46	29	29	17	15	60	60	60	60	44	37	52	52	72	73	59	49
Conical Density																		
Large Area																		
40	112	112	80	79	50	44	147	147	147	148	107	92	123	123	199	199	163	139
20	54	54	35	35	21	18	85	85	85	86	62	53	72	72	107	107	86	73
Small Area																		
40	117	117	88	88	56	49	117	117	118	118	86	74	98	98	175	174	145	123
20	61	61	40	40	24	21	72	72	72	72	52	45	60	60	96	96	79	67

Table A.2: Table of the average expected and observed detection data for the deer for all designs; regular, min(n,r) and $Q_{\bar{p}_m}$ and the various density surfaces; homogeneous density, gradient in density and conical density. $E(n)$ and n are the expected number of captures and observed captures respectively. $E(r)$ and r are the expected number of recaptures and observed recaptures respectively. $E(m)$ and m are the expected number of movements and observed movements respectively. All values were rounded to the nearest whole number.

	Regular						*min(n, r)*						$Q_{\bar{p}_m}$					
	E(n)	**n**	**E(r)**	**r**	**E(m)**	**m**	**E(n)**	**n**	**E(r)**	**r**	**E(m)**	**m**	**E(n)**	**n**	**E(r)**	**r**	**E(m)**	**m**
Homogeneous Density																		
Large Area																		
40	107	107	1290	1291	783	238	258	258	1139	1138	229	87	189	188	1209	1208	495	156
20	61	61	638	636	367	111	165	165	534	535	13	7	96	96	603	603	255	76
Small Area																		
40	61	61	638	639	367	112	129	129	569	570	112	43	93	93	605	605	262	79
20	61	61	638	639	367	112	129	129	569	570	112	43	93	93	605	605	262	79
Gradient in Density																		
Large Area																		
40	111	112	1339	1341	809	247	270	271	1351	1353	363	129	221	221	1779	1781	885	272
20	65	65	665	665	379	116	194	195	739	740	81	35	141	141	1038	1039	497	150
Small Area																		
40	110	110	1326	1325	802	244	154	155	1370	1370	737	221	148	148	1422	1422	809	239
20	63	63	649	648	369	112	133	133	646	646	163	59	107	107	853	854	439	129
Conical Density																		
Large Area																		
40	139	139	1744	1742	1065	323	286	286	1466	1467	383	145	234	234	2000	2001	980	314
20	77	77	817	817	468	143	206	206	812	812	92	44	153	153	1204	1203	569	179
Small Area																		
40	134	134	1876	1874	1178	355	164	163	1458	1455	734	232	156	156	1601	1604	884	274
20	83	83	928	927	539	163	142	142	722	723	175	69	116	116	973	974	480	149

Table A.3: Table of the average expected and observed detection data for the rabbit for all designs; regular, min(n,r) and $Q_{\bar{p}_m}$ and the various density surfaces; homogeneous density, gradient in density and conical density. $E(n)$ and n are the expected number of captures and observed captures respectively. $E(r)$ and r are the expected number of recaptures and observed recaptures respectively. $E(m)$ and m are the expected number of movements and observed movements respectively. All values were rounded to the nearest whole number.

	Regular						*min(n, r)*						$Q_{\bar{p}_m}$					
	E(n)	**n**	**E(r)**	**r**	**E(m)**	**m**	**E(n)**	**n**	**E(r)**	**r**	**E(m)**	**m**	**E(n)**	**n**	**E(r)**	**r**	**E(m)**	**m**
Homogeneous Density																		
Large Area																		
40	26	26	169	169	103	46	58	58	138	138	28	14	40	39	156	156	73	32
20	15	15	83	83	48	21	35	35	63	63	1	1	20	20	77	78	37	16
Small Area																		
40	15	15	83	83	48	21	29	29	69	69	13	7	20	20	78	78	38	16
20	15	15	83	83	48	21	29	29	69	69	13	7	20	20	78	78	38	16
Gradient in Density																		
Large Area																		
40	28	28	179	180	109	49	62	62	172	172	50	24	49	49	246	245	137	59
20	16	16	89	89	51	23	43	43	94	94	13	7	31	31	139	140	74	32
Small Area																		
40	27	27	174	173	106	47	37	37	175	175	93	41	35	35	195	195	117	50
20	15	15	84	84	48	22	30	30	82	82	22	11	24	24	116	117	65	28
Conical Density																		
Large Area																		
40	34	34	229	229	141	62	66	66	190	190	54	28	53	53	282	282	157	70
20	19	19	106	106	61	27	46	46	105	105	16	9	34	34	165	165	87	39
Small Area																		
40	34	34	248	248	156	69	40	40	193	193	99	45	38	38	224	224	132	58
20	20	20	121	121	71	31	33	33	93	93	25	13	26	26	137	136	75	33

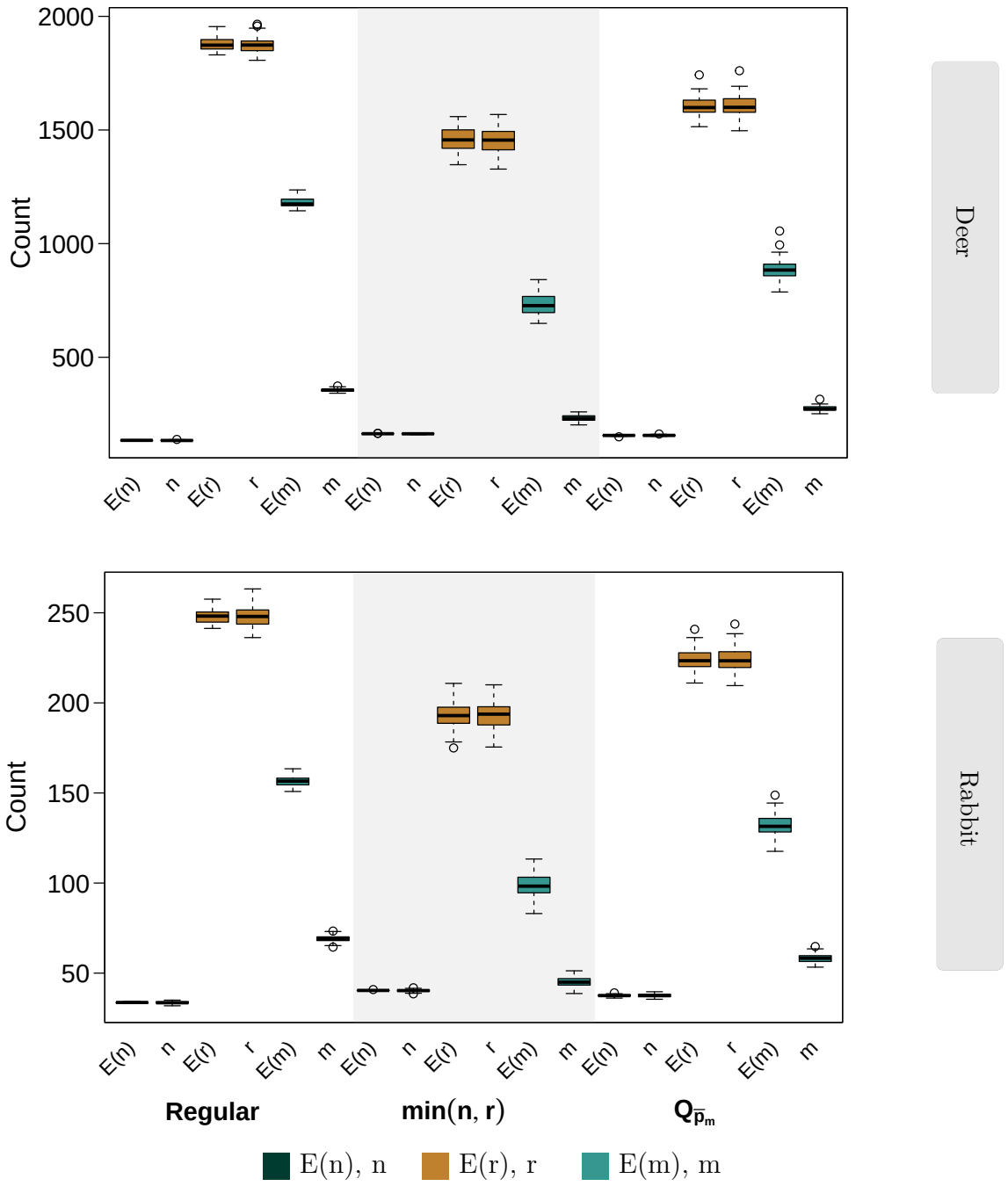

Figure A.1: Boxplots of the expected and observed detection data for the leopard for 100 realisations of each design; regular, min(n,r) and $Q_{\bar{p}_m}$, prior to simulation, for a small area with 40 traps for the deer and the rabbit for the conical density surface.

B Evaluation of Density Estimates

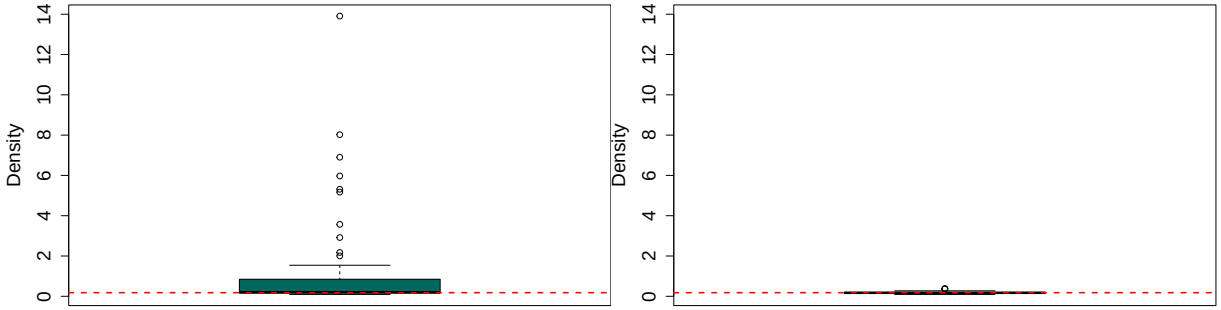

Figure B.1: Boxplots of the homogeneous density estimates for the rabbit using the large area, 20 traps, and the min(n,r) criteria. The left hand side plot shows the estimates of density for the rabbit from all 100 simulated data sets. The right hand side plot shows the estimates for the density using only strong data (67%) from the simulations. The horizontal dashedred line shows the actual density value of 0.18 rabbits per ha.

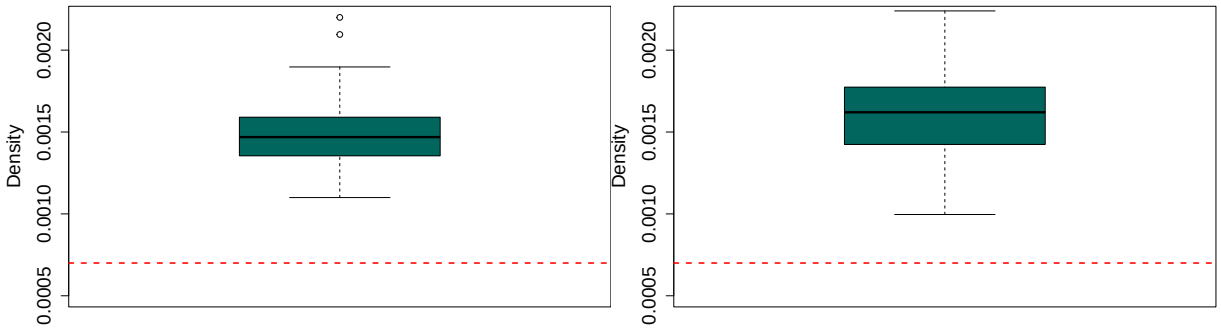

Figure B.2: Boxplots of the conical density estimates for a leopard using the large area and 20 traps. The left hand side plot shows the estimates of density for the leopard from 100 simulated data sets from the min(n,r) designs. The right hand side plot shows the estimates for the density from 100 simulated data sets from the $Q_{\bar{p}_m}$ designs. The horizontal dashed red line shows the actual density value of 0.0007 leopards per ha.

Table B.1: Table presenting the average percent relative bias, percent relative standard error, and CV($\hat{D}$) approximation for the density estimate for the leopard; 0.7 leopards per 10 km^2, compared across the three grid designs; regular, min(n,r) and $Q_{\bar{p}_m}$ and the various density surfaces; homogeneous density, gradient in density and conical density.

	Regular			min(n,r)			$Q_{\bar{p}_m}$		
	RB	**RSE**	**CV($\hat{D}$)**	**RB**	**RSE**	**CV($\hat{D}$)**	**RB**	**RSE**	**CV($\hat{D}$)**
Homogeneous Density									
Large Area									
40	0.58	13.36	13.06	0.37	15.13	11.85	0.93	14.62	12.15
20	6.48	19.65	18.93	-0.54	24.87	16.75	6.97	24.07	17.43
Small Area									
40	-0.59	13.37	13.06	0.19	13.97	11.85	1.71	14.06	12.71
20	6.37	19.61	18.93	6.87	21.59	16.75	2.73	21.97	17.75
Gradient in Density									
Large Area									
40	0.48	13.54	12.85	60.99	10.87	8.97	69.18	10.76	9.70
20	3.71	20.56	18.57	92.71	15.57	11.91	88.82	15.73	12.83
Small Area									
40	4.11	12.95	12.80	32.51	12.23	10.03	42.91	11.91	10.86
20	5.51	20.09	18.58	63.10	16.79	12.94	72.19	16.74	13.93
Conical Density									
Large Area									
40	4.74	18.62	11.21	78.26	10.03	8.26	100.10	9.74	9.02
20	6.20	25.74	16.91	111.58	13.00	10.83	131.04	13.61	11.75
Small Area									
40	8.21	19.49	10.68	46.29	11.24	9.25	64.97	10.80	10.11
20	8.63	30.01	15.75	81.25	13.91	11.80	94.95	15.00	12.89

Table B.2: Table presenting the average percent relative bias, percent relative standard error, and CV($\hat{D}$) approximation for the density estimate for the deer; 39 deers per 10 km^2, compared across the three grid designs; regular, min(n,r) and $Q_{\bar{p}_m}$ and the various density surfaces; homogeneous density, gradient in density and conical density.

	Regular			min(n,r)			$Q_{\bar{p}_m}$		
	RB	**RSE**	**CV($\hat{D}$)**	**RB**	**RSE**	**CV($\hat{D}$)**	**RB**	**RSE**	**CV($\hat{D}$)**
Homogeneous Density									
Large Area									
40	-0.41	9.78	9.68	0.43	6.66	6.23	0.20	7.50	7.28
20	-0.05	13.07	12.82	1.41	11.18	7.79	-1.01	10.87	10.20
Small Area									
40	-0.57	9.79	12.82	0.83	8.15	8.81	0.11	8.36	10.34
20	-0.07	13.09	12.82	0.53	9.42	8.81	0.90	10.74	10.34
Gradient in Density									
Large Area									
40	-44.60	27.67	9.47	-44.29	12.68	6.08	-33.32	18.55	6.72
20	-44.78	47.37	12.44	-35.21	18.60	7.17	-18.41	33.99	8.43
Small Area									
40	-46.44	21.44	9.52	-46.31	15.62	8.05	-44.75	16.62	8.22
20	-46.60	35.06	12.58	-45.18	18.10	8.68	-32.57	26.58	9.65
Conical Density									
Large Area									
40	11.94	20.22	8.48	-42.53	12.28	5.92	-30.26	17.12	6.54
20	21.59	32.07	11.38	-34.35	17.56	6.97	-17.23	30.54	8.09
Small Area									
40	9.30	20.05	8.63	-43.06	15.18	7.82	-43.76	16.43	8.01
20	9.71	28.30	10.98	-42.31	17.94	8.40	-27.07	24.70	9.29

Table B.3: Table presenting the average percent relative bias, percent relative standard error, and CV($\hat{D}$) approximation for the density estimate for the rabbit; 180 rabbits per 10 km^2, compared across the three grid designs; regular, min(n,r) and $Q_{\bar{p}_m}$ and the various density surfaces; homogeneous density, gradient in density and conical density.

	Regular			min(n,r)			$Q_{\bar{p}_m}$		
	RB	**RSE**	**CV($\widehat{D}$)**	**RB**	**RSE**	**CV($\widehat{D}$)**	**RB**	**RSE**	**CV($\widehat{D}$)**
Homogeneous Density									
Large Area									
1	0.12	20.14	19.56	3.02	15.33	13.17	4.05	16.61	15.89
2	3.10	27.71	26.17	0.24	62.08	16.89	4.66	24.69	22.22
Small Area									
3	0.85	20.24	26.17	0.21	17.06	18.57	2.05	17.56	22.62
4	5.49	27.26	26.17	2.95	22.05	18.57	2.60	24.88	22.62
Gradient in Density									
Large Area									
40	-36.24	64.99	18.97	-41.04	28.37	12.72	-19.08	46.74	14.27
20	-30.20	142.97	25.05	-30.86	46.13	15.26	7.72	96.99	18.01
Small Area									
40	-43.49	47.39	19.26	-41.97	33.10	16.37	-37.50	38.01	16.96
20	-33.90	88.52	25.73	-35.26	41.23	18.16	-26.48	71.71	20.40
Conical Density									
Large Area									
40	21.44	46.92	17.10	-35.22	27.21	12.27	-19.83	42.36	13.74
20	50.85	83.08	23.22	-23.71	42.23	14.72	11.19	83.60	17.08
Small Area									
40	12.44	44.70	17.25	-41.84	32.65	15.73	-36.79	36.97	16.33
20	28.44	67.02	22.30	-35.24	41.11	17.36	-10.75	64.82	19.48

C R-Code

```r
library(secrdesign)
library(secr)
library(foreach)
library(doParallel)
library(kableExtra)
library(sf)
library(terra)
library(dplyr)

# Define the species parameters and trap/grid configurations
D_vals    = c(0.0007, 0.18, 0.039) # Leopard, rabbit, deer
lam0_vals = c(0.02, 0.15, 0.28) * 45
sig_vals  = c(3000, 80, 336.4)

no_traps = c(40, 20)  # Number of traps
xreg = 5  # Regular grid x-dimension
ngrids = 100 # Number of design realisations

design_algorithms = c("regular_grid", 4, 5)
densities = c("D", "D1", "D2")

# Set up configuration combinations
configurations <- expand.grid(
  grid_size = c("large", "small"),  # large = 36x36, small = 18x27
  no_traps = no_traps,
  criterion = design_algorithms,
  density = densities
)

# Function to create random regular grids for inhomogeneous density
random_reg = function(x_traps, y_traps, x, y, no.traps, sig_val){

  n_col = (x*sig_val) / ((1/6)*sig_val)
  n_row = (y*sig_val) / ((1/6)*sig_val)
  x_min = 3*sig_val
  x_max = (x*sig_val-3*sig_val)
  y_min = 3*sig_val
  y_max = (y*sig_val-3*sig_val)

  # create raster for mask and for trap locations
  # vector data is a polygon - give coords of vertices which are linked to form shape,
  # raster is like elevation data
  mask.r <- rast(ncol = n_col, nrow = n_row, xmin = 0, xmax = x*sig_val,
                 ymin = 0, ymax = y*sig_val)  #ncol/row determine the resolution
  traplocs.r <- rast(ncol = x_traps, nrow = y_traps, xmin = x_min, xmax = x_max,
                     ymin = y_min, ymax = y_max)

  values(mask.r) <- 1:ncell(mask.r) # numbers each cell, seems needed
  values(traplocs.r) <- 1:ncell(traplocs.r)

  # create polygon
  msk.p1 <- rbind(c(0, 0), c(x*sig_val, 0), c(x*sig_val, y*sig_val),
                  c(0, y*sig_val), c(0,0))
```

```
  mask.v <- vect(msk.p1, type = "polygons") # creates a vector from polygon

  traplocs.p1 <- rbind(c(x_min, y_min), c(x_max, y_min), c(x_max, y_max),
                       c(x_min, y_max), c(x_min,y_min))
  traplocs.v <- vect(traplocs.p1, type = "polygons")

  # create mask and traps obj
  # get coordinates
  msk.coords <- crds(mask.r, df = T)
  traploc.coords <- crds(traplocs.r, df=T)

  mask <- read.mask(data = msk.coords, spacing = (1/6)*sig_val)
  trap.locs <- read.traps(data = traploc.coords, detector = "count",
                          spacing = (2/3)*sig_val)

  # grid with 2 sigma spacing
  locs <- st_as_sf(traplocs.v)


  grid.design <- function(poly, grid.size, nT){

    # place a grid over the area, with cells 2 * sigma apart
    my_grid <- st_make_grid(poly, cellsize = grid.size, what = "centers") %>%
      st_intersection(poly)
    grid_traps <- st_coordinates(my_grid) %>% as.data.frame() %>%
      rename(x = X, y = Y)

    # choose a subset of nT of these, starting from a random trap and
    # then choosing nT nearest neighbours
    xy_rand <- grid_traps[sample(1:nrow(grid_traps), 1), ]
    grid_traps$dist2pt <- (grid_traps$x - xy_rand$x)^2 +
                          (grid_traps$y - xy_rand$y)^2
    grid_traps <- grid_traps %>%
      filter(rank(dist2pt, ties.method = "random") <= nT) %>%
      mutate(trap_id = 1)
    opt.grid <- read.traps(data = grid_traps, detector = "count")
    return(opt.grid)
  }

  reg_grid = list()

  for(i in 1:ngrids)
  {
    reg_grid[[i]] = grid.design(locs, 2*sig_val, no.traps)
  }

  return(reg_grid)
}
```

```r
# Function to create grids and mask based on configuration
set_up_grid = function(config, sig_val, D_val) {

  # Set grid size based on configuration
  if (config$grid_size == "large") {
    x_traps = 36
    y_traps = 36
    x = 30
    y = 30
  } else {
    x_traps = 18
    y_traps = 27
    x = 18
    y = 24
  }

  # If the density is homogeneous
  if(config$density == "D") {

    # Create the regular grid, all possible trap locations and the mask
    reg_grid = replicate(ngrids, make.grid(nx = xreg,
                                           ny = (config$no_traps / 20),
                                           spacing = (2 * sig_val),
                                           detector = 'count'),
                         simplify = FALSE)
    alltrps = make.grid(nx = x_traps, ny = y_traps, spacing = (2/3) * sig_val,
                        detector = 'count')
    msk = make.mask(alltrps, type = 'traprect', spacing = 1/6 * sig_val,
                    buffer = 3 * sig_val)

    return(list(reg_grid = reg_grid, alltrps = alltrps, msk = msk, D = D_val))

  } else { # If the density is inhomogeneous

    origin_x = sig_val * 3  # To ensure all coordinates stay positive for inhomo
    origin_y = sig_val * 3

    reg_grid = random_reg(x_traps, y_traps, x, y, config$no_traps, sig_val)
    alltrps = make.grid(nx = x_traps, ny = y_traps, spacing = (2/3)*sig_val,
                        detector = 'count', originxy = c(origin_x, origin_y))
    msk = make.mask(alltrps, type = 'traprect', spacing = (1/6)*sig_val,
                    buffer = 3*sig_val)

    # For the gradient in density
    if(config$density == "D1") {
      D_grad = exp(0.5 * as.numeric(scale(sqrt(msk$x * msk$y))))
      # standardising so that the average is the same as D_val
      D_grad = D_grad/sum(D_grad) * (D_val * nrow(msk))
      covariates(msk) = data.frame(D1 = D_grad)
      poplist = list(list(model2D = "IHP", D = "D1", core = msk))

      return(list(reg_grid = reg_grid, alltrps = alltrps, msk = msk,
                  D = D_grad, poplist = poplist))
```

```
    # For the conical density
    } else if(config$density == "D2") {
      D_center = exp(-0.5 * as.numeric(scale(sqrt((msk$x-mean(msk$x))^2 +
                                                  (msk$y-mean(msk$y))^2))))
      D_center = D_center/sum(D_center) * (D_val * nrow(msk))
      covariates(msk) = data.frame(D1 = D_center)
      poplist = list(list(model2D = "IHP", D = "D1", core = msk))

      return(list(reg_grid = reg_grid, alltrps = alltrps, msk = msk,
                  D = D_center, poplist = poplist))
    }
  }
}

# Function to run GA optimization based on configuration
run_GA = function(grids, config, lam0_val, sig_val) {
  num_cores = detectCores() - 1  # Parallelize for efficiency
  cl = makeCluster(num_cores)
  registerDoParallel(cl)

  optimal_designs = foreach(i = 1:ngrids, .packages = c("secrdesign")) %dopar% {
    opt_min = GAoptim(grids$msk, grids$alltrps, ntraps = config$no_traps,
              detectpar = list(lambda0 = lam0_val, sigma = sig_val),
              criterion = as.numeric(as.character(config$criterion)),
              detectfn = 'HHN', D = grids$D,
              noccasions = 1, ngen = 50, verbose = 0)
    return(opt_min$optimaltraps)
  }

  stopCluster(cl)
  return(optimal_designs)
}

# Function to run simulations based on design and population type
run_simulation = function(config, design, grids, D_val, lam0_val, sig_val) {
  scen = make.scenarios(trapsindex = 1:length(design), noccasions = 1,
                        nrepeats = 1,
                        D = D_val, lambda0 = lam0_val, sigma = sig_val,
                        detectfn = 14, recapfactor = 1, popindex = 1,
                        detindex = 1, fitindex = 1, crosstraps = FALSE)

  # Do different simulations dependent on if its homogeneous density or not
  if (config$density == "D") {
    sim_Ffit = run.scenarios(nrepl = 100, scenarios = scen, trapset = design,
                             fit = FALSE, maskset = grids$msk,
                             fit.args = list(detectfn = "HHN"),
                             det.args = list(savepopn = TRUE),
                             extractfn = identity)
    sim = run.scenarios(nrepl = 1, scenarios = scen, trapset = design,
                        fit = TRUE, maskset = grids$msk,
                        fit.args = list(detectfn = "HHN"))
  } else {
    sim_Ffit = run.scenarios(nrepl = 100, scenarios = scen, trapset = design,
```

```
                              fit = FALSE, maskset = grids$msk,
                              fit.args = list(detectfn = "HHN"),
                              pop.args = grids$poplist,
                              det.args = list(savepopn = TRUE),
                              extractfn = identity)
    sim = run.scenarios(nrepl = 1, scenarios = scen, trapset = design,
                        fit = TRUE, maskset = grids$msk,
                        fit.args = list(model = list(D ~ D1), detectfn = "HHN"),
                        pop.args = grids$poplist)
  }

  return(list(sim_Ffit = sim_Ffit, sim = sim))
}


# Function to calculate expected and obs detection data for each configuration
get_enrm = function(config, grids, designs, sim, lam0_val, sig_val){

  sum = summary(sim)

  n = c()
  r = c()
  nmov = c()
  enrm = c()
  En = c()
  Er = c()
  Em = c()

  for(i in 1:ngrids) {
    # Extracting observed data
    n[i] = sum$OUTPUT[[i]][1,2]
    r[i]   = sum$OUTPUT[[i]][2,2]
    nmov[i]= sum$OUTPUT[[i]][3,2]

    # Calculating expected data
    enrm[[i]] = Enrm(grids$D, designs[[i]], grids$msk, list(lambda0 = lam0_val,
                        sigma = sig_val), noccasions = 1)

    En[i] = enrm[[i]][1]
    Er[i] = enrm[[i]][2]
    Em[i] = enrm[[i]][3]

  }
  # Calculating approximate CV(D)
  CV = (1 / sqrt(min(mean(En), mean(Er)))) * 100

  return(list(n = n, r = r, nmov = nmov, En = En, Er = Er, Em = Em, CV = CV))
}

get_stats = function(sim, D_val) {
  D_est = c()
  se.est = c()
```

```
  for(i in 1:ngrids){
    D_est = sim$output[[i]][[1]][1,2]
    se.est = sim$output[[i]][[1]][1,3]
  }

  avg_RB = mean((D_est- D_val) / D_val) *100
  avg_RSE = mean(se.est / D_val)

  return(list(D_est = D_est, se.est = se.est, avg_RB = avg_RB,
              avg_RSE = avg_RSE))
}

# Run all configurations for each case study
all_designs = list()
all_sims = list()
all_enrm = list()
all_stats = list()

for (j in 1:3) {

  designs = list()
  sims = list()
  enrm = list()
  stats = list()

  for (i in 1:nrow(configurations)) {
    config = configurations[i, ]
    grids = set_up_grid(config, sig_vals[j], D_vals[j])

    # Get designs
    if (config$criterion == 4 || config$criterion == 5) {
      designs[[i]] = run_GA(grids, config, lam0_vals[j], sig_vals[j])
    } else {
      designs[[i]] = grids$reg_grid  # Use regular grid for non-GA options
    }

    # Run simulations
    sim_results = run_simulation(config, designs[[i]], grids, D_vals[j],
                                 lam0_vals[j], sig_vals[j])
    sims[[i]] = sim_results

    # Calculate results
    enrm[[i]] = get_enrm(config, grids, designs[[i]], sims[[i]]$sim_Ffit,
                         lam0_vals[j], sig_vals[j])
    stats[[i]] = get_stats(sims[[i]]$sim, D_vals[j])
    cat("Species", j, "configuration", i, "done.\n")
  }

  all_designs[[j]] = designs
  all_sims[[j]] = sims
  all_enrm[[j]] = enrm
  all_stats[[j]] = stats
}
```

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